Note: an asterisk (*) in front of the item indicates that the revised version of the given item is available in "CMESpring2013Pilot_ItemList_revised_[2014.07.15]". Description of the item data is in Appendix A.

1. *Bernoulli – Facet IV – Generating Explanations / Drawing Conclusions

- *Reasoning Skills*

  Learners must understand the underlying theoretical cause for the evidence (phenomenon).

  Task Analysis:

  1. Understand the principle/theory
  2. Understand and evaluate the explanation
  3. Reflect on the strengths and weaknesses of the explanation

     - Assess the strength of the explanation

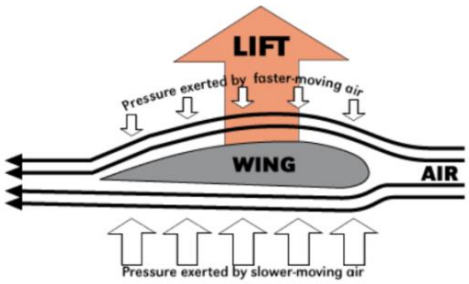
     - Identify inconsistencies
  4. Make a selection

- *Description*

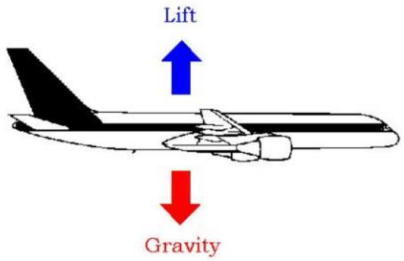
  Engineers use scientific principles to design the shape of airplane wing. Bernoulli's Principle suggests that fast flowing air creates an area of low pressure while slow moving air creates relative high pressure. How could you use this principle to explain how an airplane flies? Which statement below is true?



A. The shape of an airplane's wing causes air to travel over the top of the wing faster than under the wing, which makes the air pressure above the wing less than the pressure beneath the wing.



B. In order to fly, the force of lift has to exceed the force of gravity. Therefore, there is jet equipment under airplane wings to push the airplane up.



C. Because the shape of airplane wings is like birds' wings and much bigger than the aircraft itself, the airplane can fly easily. In addition, there is jet equipment behind the airplane wings to push the airplane forward.



- *Rubric*

Score	Response
1	A
0	B or C

- *IRT Item Statistics*

	Estimate	SE	Outfit	Infit
Middle Schoolers (MS)	0.62	0.45	0.80	0.83
High Schoolers (HS)	0.12	0.27	1.11	1.12
Adults (AD)	0.63	0.30	1.06	1.02
Combined Sample (MS, HS, & AD)	0.28	0.18	0.98	1.01
Gender DIF (Male) for Combined Sample	-0.37	0.23	0.79	1.03
Language DIF (English) for Combined Sample	-0.28	0.25	1.00	1.01
Race DIF (White) for Combined Sample	-0.01	0.14	1.13	1.09
Race DIF (Black) for Combined Sample	-0.50	0.26	1.15	1.21
Race DIF (Other) for Combined Sample	0.51	0.30	0.89	0.96
MotherEd DIF (below HS) for Combined Sample	-0.65	0.28	2.70	2.22
MotherEd DIF (HS) for Combined Sample	0.04	0.20	0.65	0.79
MotherEd DIF (College) for Combined Sample	0.36	0.16	1.27	1.17
MotherEd DIF (Graduate) for Combined Sample	0.25	0.38	1.07	0.97

2. Birth Rate – Facet III – Reasoning from Evidence to Conclusions

- *Reasoning Skills*

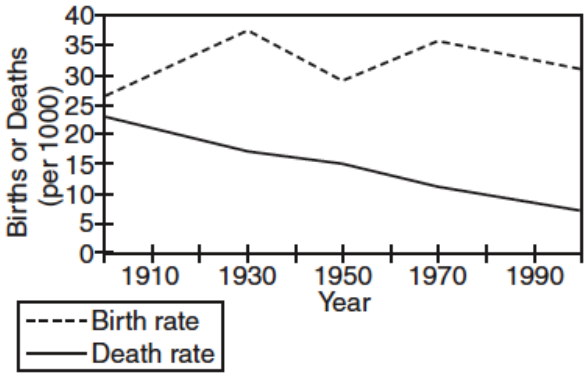
  Learners must draw correct conclusions based on a graphical data representation.

  Task Analysis:

  1. Interpret Graph
  2. Draw Conclusion

- *Description*

  The graph below shows the birth rate and death rate for a population during the 1900's.



  Which of the following statements is true based on the data in the graph?

  A. From 1910 to 1990, the population increased.
  B. From 1910 to 1990, the population decreased.
  C. From 1910 to 1990, the population stayed the same.
  D. From 1910 to 1990, the population increased until 1930, then decreased.

- *Rubric*

Score	Response
1	A
0	B, C, or D

- *IRT Item Statistics*

	Estimate	SE	Outfit	Infit
Middle Schoolers (MS)	1.41	0.30	0.86	0.91
High Schoolers (HS)	0.45	0.19	1.02	1.01
Adults (AD)	0.33	0.24	0.89	0.97
Combined Sample (MS, HS, & AD)	0.49	0.12	0.86**	0.88**
Gender DIF (Male) for Combined Sample	-0.15	0.14	0.78	0.84
Language DIF (English) for Combined Sample	-0.08	0.17	0.81	0.86
Race DIF (White) for Combined Sample	-0.35	0.12	0.93	0.96
Race DIF (Black) for Combined Sample	0.37	0.20	0.90	0.93
Race DIF (Other) for Combined Sample	-0.02	0.24	0.62	0.70
MotherEd DIF (below HS) for Combined Sample	-0.67	0.24	0.46	0.64
MotherEd DIF (HS) for Combined Sample	0.27	0.17	0.71	0.79
MotherEd DIF (College) for Combined Sample	0.54	0.13	0.82	0.90
MotherEd DIF (Graduate) for Combined Sample	-0.14	0.32	1.39	1.09

3. *Brown Trout – Facet III – Reasoning from Evidence to Conclusions

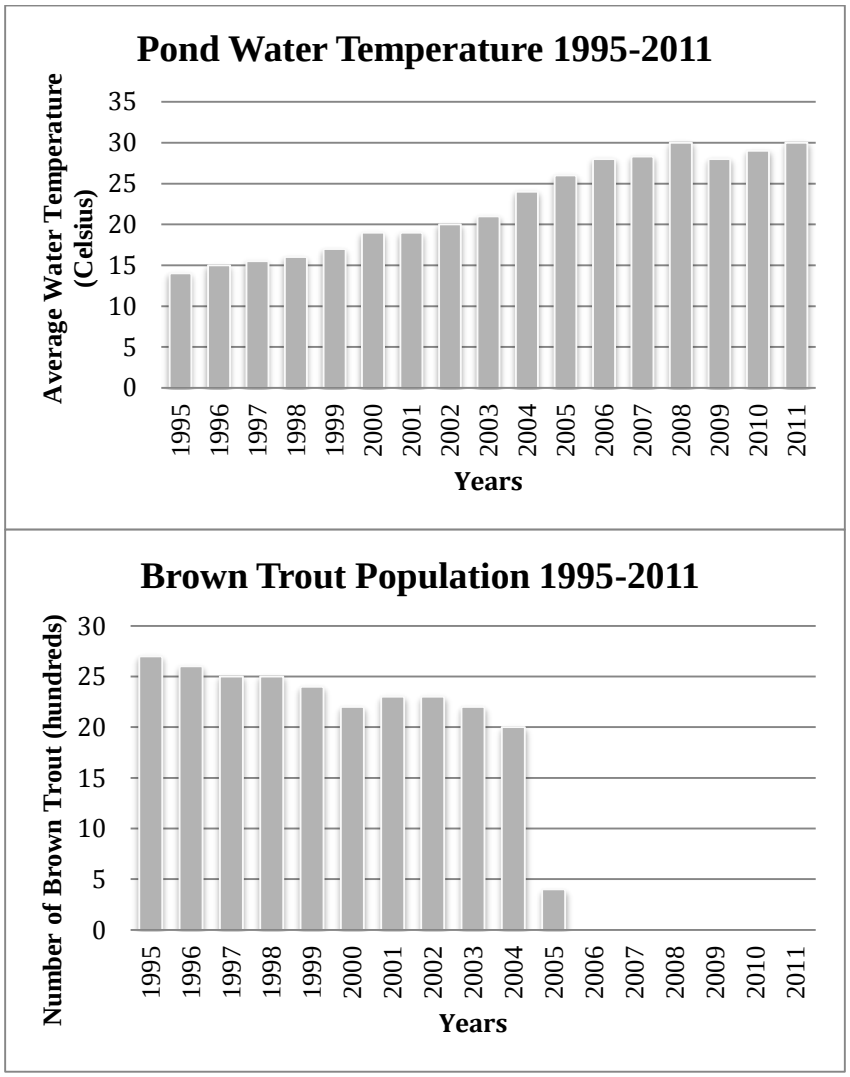
- *Reasoning Skills*

  Learners must draw justified inferences by comparing independent or dependent relationships.

  Task Analysis:

  1. Interpret graph.
  2. Compare relationships
  3. Induce a general statement about the relationships among data that summarize the evidence

- *Description*



Use the data in the graphs above to answer the following question: How is the Brown Trout population affected by the pond water temperature?

The Brown Trout population _____.

A. is not affected by the pond water temperature.

B. deceases after the pond water reaches a certain temperature.

C. tends to increase as the pond water temperature increases.

D. is highest when the temperature is the highest.

- *Rubric*

Score	Response
1	B
0	A, C, or D

- *IRT Item Statistics*

	Estimate	SE	Outfit	Infit
Middle Schoolers (MS)	-0.48	0.42	0.86	0.91
High Schoolers (HS)	-0.78	0.29	0.78	0.90
Adults (AD)	-2.07	0.97	0.59**	0.97
Combined Sample (MS, HS, & AD)	-0.85	0.22	0.50**	0.82
Gender DIF (Male) for Combined Sample	-0.01	0.28	0.48	0.76
Language DIF (English) for Combined Sample	-0.05	0.27	0.47	0.85
Race DIF (White) for Combined Sample	-0.87	0.16	0.34	0.91
Race DIF (Black) for Combined Sample	0.76	0.27	0.71	0.79
Race DIF (Other) for Combined Sample	0.12	0.32	0.66	0.95
MotherEd DIF (below HS) for Combined Sample	1.09	0.33	.	.
MotherEd DIF (HS) for Combined Sample	0.10	0.22	0.39	0.67
MotherEd DIF (College) for Combined Sample	-0.29	0.18	0.52	0.89
MotherEd DIF (Graduate) for Combined Sample	-0.90	0.44	0.76	1.12

4. *Cans of Water – Facet III – Reasoning from Evidence to Conclusions

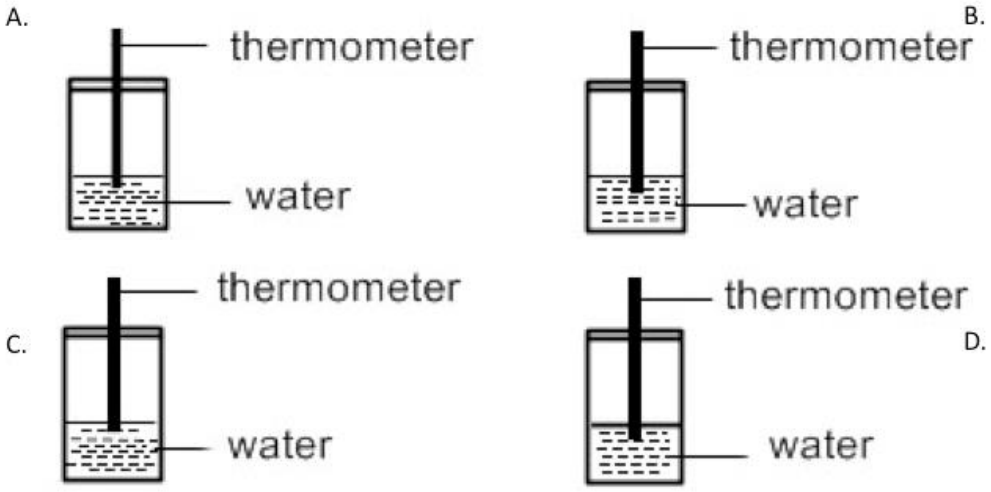
- *Reasoning Skills*
  Learners must interpret the results of an investigation to induce a general statement about the results.

  Task Analysis:

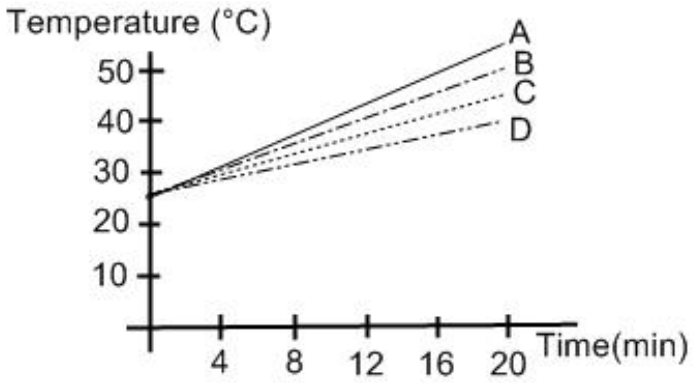
  1. Understand the experimental design and goal
  2. Understand the results of the investigation
  3. Interpret one or more graphs
  4. Recognize patterns in the data
  5. Draw a valid, evidence-based conclusion

- *Description*

Equal amount of tap water was poured into four cans of identical shape and size. The cans were made of different materials A, B, C and D. A thermometer was inserted into each can. The cans were then placed in an open field on a sunny day.



The graph below shows the temperature change of the water in each can after 20 minutes.



Based on the graph, which of the materials A, B, C or D should be used to make a box to prevent ice cubes from melting too quickly?

A. A

B. B

C. C

D. D

- *Rubric*

Score	Response
1	D
0	A, B, or C

- *IRT Item Statistics*

	Estimate	SE	Outfit	Infit
Middle Schoolers (MS)	-0.51	0.34	0.75	0.82
High Schoolers (HS)	-0.14	0.36	0.85	0.91
Adults (AD)	-0.45	0.45	0.85	0.98
Combined Sample (MS, HS, & AD)	-0.40	0.21	0.63**	0.83
Gender DIF (Male) for Combined Sample	0.30	0.26	0.64	0.92
Language DIF (English) for Combined Sample	-0.21	0.25	0.66	0.89
Race DIF (White) for Combined Sample	0.12	0.15	0.71	0.91
Race DIF (Black) for Combined Sample	0.10	0.28	0.79	0.91
Race DIF (Other) for Combined Sample	-0.22	0.32	0.68	0.87
MotherEd DIF (below HS) for Combined Sample	0.61	0.31	.	.
MotherEd DIF (HS) for Combined Sample	-0.53	0.22	0.68	1.03
MotherEd DIF (College) for Combined Sample	0.15	0.17	0.69	0.85
MotherEd DIF (Graduate) for Combined Sample	-0.23	0.41	0.50	0.83

5. *City Birds – Facet V – Coordinating Theory & Evidence

- *Reasoning Skills*

  In this item, learners must coordinate theory and evidence in such a way so as to draw inferences about causal relationships.

  Task Analysis:

  1. Understand the theory
  2. Interpret Evidence
  3. Coordinate evidence and conclusions

- *Description*

  The table below shows the number of birds of a particular color living in a northern city where the amount of snow per year was declining.

	Year 1	Year 5	Year 10
White birds	100	50	25
Gray birds	50	75	100
Black birds	50	50	50

Evolutionary theory suggests that populations of organisms are changing over generations as they adapt to their environments. How could you use evolutionary theory to explain the data above? Circle the statement that provides the best explanation.

A. The number of white birds declined because there was less snow and the birds were no longer camouflaged, and more were killed by predators.

B. The number of white birds declined because there was less snow and the birds no longer needed the white feathers.

C. The number of gray birds increased because the gray color was warmer.

D. The number of black birds stayed about the same because color has no impact on getting killed by predators.

- *Rubric*

Score	Response
1	A
0	B, C, or D

- *IRT Item Statistics*

	Estimate	SE	Outfit	Infit
Middle Schoolers (MS)	-0.10	0.21	1.01	1.03
High Schoolers (HS)	-0.10	0.16	1.02	1.01
Adults (AD)	0.57	0.18	1.04	1.02
Combined Sample (MS, HS, & AD)	0.05	0.11	1.08	1.04
Gender DIF (Male) for Combined Sample	0.03	0.12	1.07	1.03
Language DIF (English) for Combined Sample	0.07	0.14	1.09	1.05
Race DIF (White) for Combined Sample	0.23	0.11	1.17	1.07
Race DIF (Black) for Combined Sample	-0.36	0.18	0.97	0.96
Race DIF (Other) for Combined Sample	0.13	0.21	1.39	1.23
MotherEd DIF (below HS) for Combined Sample	-1.66	0.21	0.31	0.77
MotherEd DIF (HS) for Combined Sample	0.42	0.15	1.39	1.18
MotherEd DIF (College) for Combined Sample	0.51	0.12	0.96	1.02
MotherEd DIF (Graduate) for Combined Sample	0.73	0.28	1.28	1.13

6. Crickets – Facet II – Hypothesis Testing

- *Reasoning Skills*

  Learners must design a valid test by manipulating all relevant variables in order that empirical evidence may be produced that will allow them to answer the question.

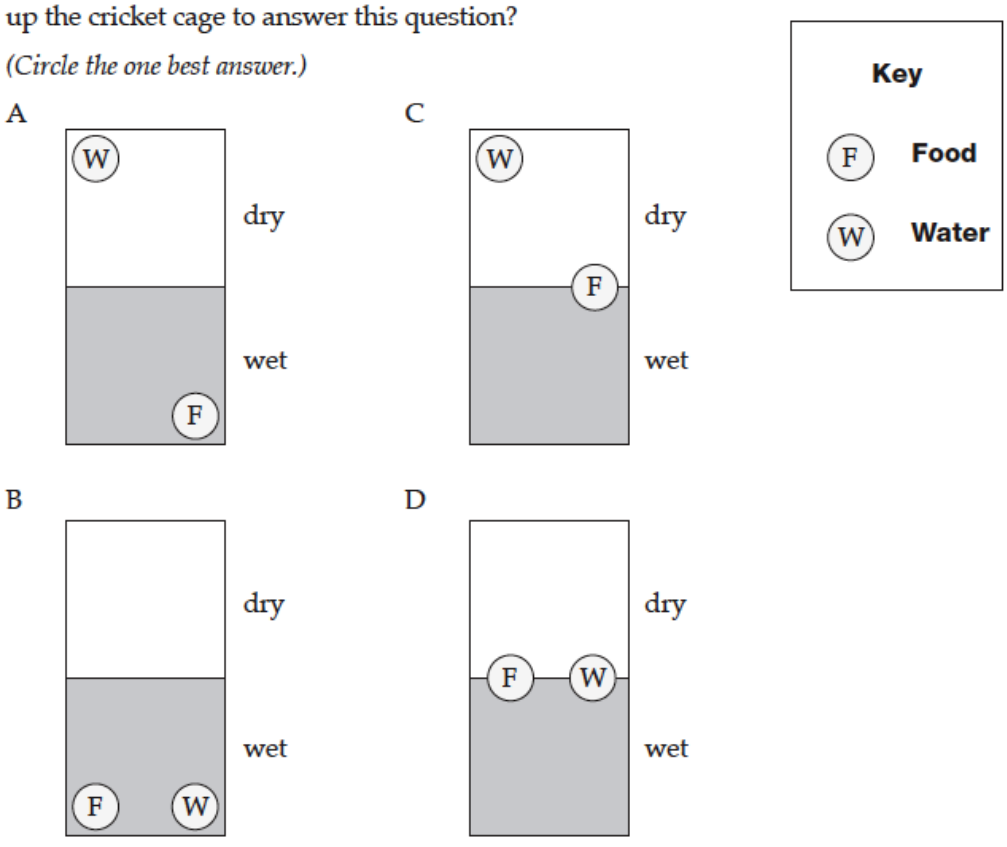
  Task Analysis:

  1. Understand experiment goal
  2. Understand the experiment design
  3. Identify relevant variables
  4. Use controlling variables strategy to determine effect on the dependent variable

- *Description*

Kate wants to know if crickets prefer wet or dry environments. How should she set up the cricket cage to answer this question?

(Circle the one best answer.)



- *Rubric*

Score	Response
1	D
0	A, B, or C

- *IRT Item Statistics*

	Estimate	SE	Outfit	Infit
Middle Schoolers (MS)	-1.06	0.45	1.08	1.08
High Schoolers (HS)	-0.43	0.28	1.11	1.10
Adults (AD)	-1.02	0.58	0.72**	0.98
Combined Sample (MS, HS, & AD)	-0.73	0.21	0.80**	1.03
Gender DIF (Male) for Combined Sample	0.29	0.25	0.95	1.05
Language DIF (English) for Combined Sample	0.70	0.34	0.68	0.95
Race DIF (White) for Combined Sample	0.07	0.15	0.69	0.99
Race DIF (Black) for Combined Sample	0.22	0.27	0.86	0.94
Race DIF (Other) for Combined Sample	-0.29	0.31	0.82	1.14
MotherEd DIF (below HS) for Combined Sample	0.59	0.29	0.44	0.49
MotherEd DIF (HS) for Combined Sample	-0.03	0.21	0.81	1.07
MotherEd DIF (College) for Combined Sample	-0.42	0.17	0.93	1.12
MotherEd DIF (Graduate) for Combined Sample	-0.14	0.40	0.53	0.81

7. *Evolution – Facet IV – Generating Explanations / Drawing Conclusions

- *Reasoning Skills*

  Learners must coordinate theory and evidence to explain phenomena. They must seek an underlying theoretical cause for the evidence.

  Task Analysis:

  1. Understand theory
  2. Explain evidence/phenomenon
  3. Construct (select) a theory based argument that is aligned with theory and accounts for experimental outcomes and/or observed phenomena

- *Description*

  Evolutionary theory suggests that populations of organisms are changing over generations as they adapt to their environments. For example, the development of DDT-resistant corn earworms can be explained through evolutionary theory as follows:

Years ago spraying corn crops with a chemical called DDT killed almost all of the corn earworms that would infest corn crops. The few corn earworms that survived were DDT-resistant. These surviving earworms were able to reproduce more DDT-resistant earworms. Eventually, spraying with DDT killed only a few of the corn earworms.

Which of the following phenomena could be explained using evolutionary theory? Select the best answer.

A. Organisms have long life cycles

B. Over time spraying with chemicals can kill insects

C. Medicine helps people with diseases live longer

D. Over time a population of organisms could have better sight

E. Treating infections with antibiotics could lead to bacteria that are resistant to antibiotics

- *Rubric*

Score	Response
1	E
0	A, B, C or D

- *IRT Item Statistics*

	Estimate	SE	Outfit	Infit
Middle Schoolers (MS)	-0.20	0.39	0.89	0.95
High Schoolers (HS)	-0.44	0.28	0.88	0.86
Adults (AD)	-0.51	0.45	0.95	1.00
Combined Sample (MS, HS, & AD)	-0.45	0.20	0.82	0.87
Gender DIF (Male) for Combined Sample	-0.34	0.24	0.60	0.82
Language DIF (English) for Combined Sample	-0.19	0.26	0.70	0.85
Race DIF (White) for Combined Sample	-0.80	0.15	0.77	0.91
Race DIF (Black) for Combined Sample	0.80	0.26	0.97	0.99
Race DIF (Other) for Combined Sample	-0.01	0.30	0.80	0.77
MotherEd DIF (below HS) for Combined Sample	1.36	0.30	.	.
MotherEd DIF (HS) for Combined Sample	-0.28	0.21	0.61	0.77
MotherEd DIF (College) for Combined Sample	-0.48	0.17	0.76	0.85
MotherEd DIF (Graduate) for Combined Sample	-0.60	0.40	0.94	1.04

8. Falling Objects – Facet V – Coordinating Theory & Evidence

- *Reasoning Skills*

Learners must evaluate a theory in light of an experimental conclusion, reconcile new evidence with prior beliefs and (if required) revise one's theory and generate new predictions.

Task Analysis:

1. Understand the theory
2. Interpret the experiment
3. Understand and evaluate the evidence
4. Reflect on strengths and weaknesses of the theory in light of new evidence

- *Description*

Gravitational theory states the force of gravity affects all objects the same way. Mark believes that heavy objects fall faster than light objects, but he saw a video of an astronaut on the moon dropping a feather and hammer, and the hammer and feather hit the ground at same time. Mark decides to test it out himself here on Earth. He drops a feather and hammer from the same height at the same time and measures how long it takes for each object to hit the floor. He does the test several times. The hammer always reaches the floor before the feather. Which of the following statements is the best explanation?

A. The scientific theory is wrong because the feather and hammer fell at different speeds on earth.

B. The hammer fell faster on earth because the hammer is larger.

C. The objects have to be the same shape to test the theory.

D. There are other variables involved when he drops the two objects.

- *Rubric*

Score	Response
1	D
0	A, B, or C

- *IRT Item Statistics*

	Estimate	SE	Outfit	Infit
Middle Schoolers (MS)	0.46	0.17	0.90	0.91
High Schoolers (HS)	0.67	0.12	0.97	0.97
Adults (AD)	0.34	0.16	1.01	1.00
Combined Sample (MS, HS, & AD)	0.44	0.08	0.87**	0.90**
Gender DIF (Male) for Combined Sample	-0.05	0.09	0.93	0.94
Language DIF (English) for Combined Sample	-0.09	0.11	0.89	0.91
Race DIF (White) for Combined Sample	-0.06	0.08	0.94	0.93

Race DIF (Black) for Combined Sample	0.00	0.14	1.13	1.01
Race DIF (Other) for Combined Sample	0.06	0.16	0.76	0.83
MotherEd DIF (below HS) for Combined Sample	-0.01	0.17	0.95	0.98
MotherEd DIF (HS) for Combined Sample	0.26	0.11	0.84	0.84
MotherEd DIF (College) for Combined Sample	-0.06	0.09	0.97	0.95
MotherEd DIF (Graduate) for Combined Sample	-0.20	0.22	0.97	0.89

9. Flashlights – Facet I – Hypothesis Generation

- *Reasoning Skills*

  Learners must show the ability to observe the situation and clearly articulate a question that can direct an empirical investigation.

  Task Analysis:

  1. Observe the experimental situation
  2. Understand the goal
  3. Make a statement of a possible investigation based on the experimentation goals

- *Description*

  Ms. Yamada's class wanted to find out which of three flashlights was the brightest.

  Which one of the statements below gives the best approach toward finding the answer?

  (Circle the one best answer)

  A. "I can look at the flashlights and tell which is the brightest."

  B. "All the flashlights look bright to me, so none are the brightest."

  C. "We can take a vote, and everyone will vote for the flashlight he or she thinks is the brightest."

  D. "We need to find a way to measure the brightness of each flashlight."

- *Rubric*

Score	Response
1	D
0	A, B, or C
-999	(no response)

- *IRT Item Statistics*

	Estimate	SE	Outfit	Infit
Middle Schoolers (MS)	-0.92	0.19	0.81	0.92
High Schoolers (HS)	-0.84	0.14	0.86	0.92
Adults (AD)	-1.43	0.32	0.92	1.00
Combined Sample (MS, HS, & AD)	-0.95	0.10	0.69**	0.89
Gender DIF (Male) for Combined Sample	0.31	0.13	0.67	0.92
Language DIF (English) for Combined Sample	-0.10	0.13	0.74	0.93
Race DIF (White) for Combined Sample	-0.10	0.11	0.69	0.92
Race DIF (Black) for Combined Sample	-0.01	0.17	0.91	0.96
Race DIF (Other) for Combined Sample	0.11	0.20	0.57	0.85
MotherEd DIF (below HS) for Combined Sample	0.28	0.20	0.91	1.05
MotherEd DIF (HS) for Combined Sample	-0.15	0.15	0.72	0.98
MotherEd DIF (College) for Combined Sample	-0.31	0.13	0.69	0.90
MotherEd DIF (Graduate) for Combined Sample	0.18	0.28	0.73	0.82

10. Gravity – Facet IV – Generating Explanations / Drawing Conclusions

- *Reasoning Skills*

  Learners must employ theory to describe a phenomenon.

  Task Analysis:

  1. Understand the theory
  2. Identify anomalies and inconsistencies in the explanations

- *Description*

  Gravitational theory suggests that all objects on earth are pulled toward the center of the earth with the same force.

  If that theory is true, which of the following would be true?

  A. A heavy ball would fall faster than a light ball.

  B. A light ball would fall faster than a heavy ball.

  C. A heavy ball and a light ball would fall at the same rate.

- *Rubric*

Score	Response
1	C
0	A or B

- *IRT Item Statistics*

	Estimate	SE	Outfit	Infit
Middle Schoolers (MS)	-0.18	0.41	1.08	1.00
High Schoolers (HS)	0.05	0.27	1.03	1.05
Adults (AD)	0.92	0.27	1.06	1.02
Combined Sample (MS, HS, & AD)	0.24	0.18	1.16	1.09
Gender DIF (Male) for Combined Sample	-0.60	0.21	0.74	0.93
Language DIF (English) for Combined Sample	-0.02	0.24	1.06	1.03
Race DIF (White) for Combined Sample	0.58	0.14	1.05	1.00
Race DIF (Black) for Combined Sample	-0.19	0.26	1.23	1.19
Race DIF (Other) for Combined Sample	-0.39	0.30	1.77	1.12
MotherEd DIF (below HS) for Combined Sample	0.04	0.24	1.54	1.48
MotherEd DIF (HS) for Combined Sample	-0.56	0.18	1.18	1.06
MotherEd DIF (College) for Combined Sample	-0.37	0.15	1.17	1.03
MotherEd DIF (Graduate) for Combined Sample	0.89	0.34	0.88	0.95

11. KMT – Facet IV – Generating Explanations / Drawing Conclusions

- *Reasoning Skills*

  Learners must coordinate theory and evidence in such a way so as to draw inferences that account for a causal relationship.

  Task Analysis:

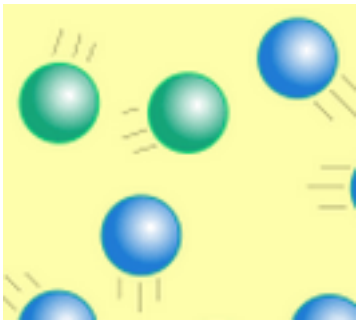
  1. Understand theory

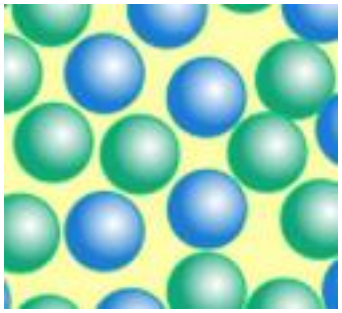
  2. Construct (select) an argument that is aligned with the theory and accounts for the experimental outcomes or observed phenomenon

- *Description*

  Kinetic molecular theory (KMT) describes the behavior of molecules. According to KMT, gases expand when heated because molecules are moving faster and therefore move further apart. For gases inside a container, the faster movement of molecules also increases the pressure exerted by the molecules on the container. In contrast, gases contract when cooled because molecules are moving slower and therefore get closer together. For gases inside a container, the slower movement of molecules also decreases the pressure exerted by the molecules on the container.

  a. In the pictures below of the same sized container, circle the picture that shows molecules of a gas at the higher temperature? Circle One.





b. At 0 degrees Celsius (freezing point of water), liquid water and ice co-exist. The liquid water molecules are closer together than ice at 0 degrees Celsius. Is this consistent with the KMT? Circle your response.

Yes or No

c. Which of the following phenomena could be explained using KMT?

(*Circle one best answer*)

A. Chip bag expands when moved from high altitude to low altitude.

B. Molecules are different sizes.

C. A balloon filled with air will expand when placed in the freezer

D. The pressure in your car's tires decreases in the winter when it is cold.

- *Rubric*

a.

Score	Response
1	Left picture
0	Right picture

b.

Score	Response
1	No
0	Yes

c.

Score	Response
1	D
0	A, B, or C

- *IRT Item Statistics*

a.

	Estimate	SE	Outfit	Infit
Middle Schoolers (MS)	-1.99	0.64	1.23	1.18
High Schoolers (HS)	-1.70	0.44	0.99	0.97
Adults (AD)	-1.41	0.70	1.02	1.01
Combined Sample (MS, HS, & AD)	-1.74	0.33	1.26**	1.09
Gender DIF (Male) for Combined Sample	-0.48	0.44	0.80	1.29
Language DIF (English) for Combined Sample	-0.21	0.37	2.14	1.10
Race DIF (White) for Combined Sample	0.02	0.17	1.26	0.99
Race DIF (Black) for Combined Sample	-0.76	0.31	1.07	1.25
Race DIF (Other) for Combined Sample	0.73	0.35	1.73	1.19
MotherEd DIF (below HS) for Combined Sample	1.13	0.33	.	.
MotherEd DIF (HS) for Combined Sample	0.18	0.23	2.43	1.21
MotherEd DIF (College) for Combined Sample	-0.34	0.20	0.61	1.02
MotherEd DIF (Graduate) for Combined Sample	-0.98	0.45	0.17	1.09

b.

	Estimate	SE	Outfit	Infit
Middle Schoolers (MS)	1.20	0.41	1.15	1.09
High Schoolers (HS)	0.60	0.27	1.28	1.19
Adults (AD)	1.57	0.23	1.01	1.00
Combined Sample (MS, HS, & AD)	1.02	0.16	1.20	1.12
Gender DIF (Male) for Combined Sample	-0.16	0.19	1.34	1.16
Language DIF (English) for Combined Sample	0.20	0.23	1.12	1.05
Race DIF (White) for Combined Sample	0.55	0.13	1.10	1.05
Race DIF (Black) for Combined Sample	-0.73	0.23	1.49	1.42
Race DIF (Other) for Combined Sample	0.18	0.27	0.96	0.90
MotherEd DIF (below HS) for Combined Sample	-1.02	0.25	2.21	1.87
MotherEd DIF (HS) for Combined Sample	0.83	0.19	1.58	1.25
MotherEd DIF (College) for Combined Sample	0.24	0.15	0.90	0.94
MotherEd DIF (Graduate) for Combined Sample	-0.05	0.34	1.29	1.20

c.

	Estimate	SE	Outfit	Infit
Middle Schoolers (MS)	-0.21	0.41	1.17	1.14
High Schoolers (HS)	0.46	0.27	0.80	0.84
Adults (AD)	0.03	0.37	0.91	0.99
Combined Sample (MS, HS, & AD)	0.11	0.18	0.78**	0.91
Gender DIF (Male) for Combined Sample	-0.09	0.22	0.53	0.77
Language DIF (English) for Combined Sample	0.39	0.26	0.77	0.87
Race DIF (White) for Combined Sample	-0.57	0.15	0.73	0.91
Race DIF (Black) for Combined Sample	0.65	0.24	0.88	0.93
Race DIF (Other) for Combined Sample	-0.08	0.28	0.82	0.98
MotherEd DIF (below HS) for Combined Sample	0.69	0.26	0.56	0.62
MotherEd DIF (HS) for Combined Sample	-0.27	0.20	0.79	0.96
MotherEd DIF (College) for Combined Sample	-0.23	0.16	0.80	0.99
MotherEd DIF (Graduate) for Combined Sample	-0.18	0.37	0.71	0.82

12. Mineral Study – Facet III – Reasoning from Evidence to Conclusions

- *Reasoning Skills*

  Learners must identify the evidence that justifies the given conclusion.

  Task Analysis:

  1. Understand the experimental design and goal

  2. Draw a valid, evidence-based conclusion

- *Description*

  A mineral supplement designed to prevent the common cold was given to two groups of people during a scientific study.

	Dosage
Group 1	$50 \frac{mg}{day}$
Group 2	$100 \frac{mg}{day}$

After eight weeks, neither group reported a case of the common cold. Which of the following would have made the design of the study more valid?

A. Test only one group with 150 mg of the supplement.

B. Give the supplement to both groups for only 6 weeks.

C. Create a third group that receives 75 mg of the supplement.

D. Create a third group that does not receive the supplement.

- *Rubric*

Score	Response
1	D
0	A, B, or C

- *IRT Item Statistics*

	Estimate	SE	Outfit	Infit
Middle Schoolers (MS)	-0.08	0.29	0.95	0.96
High Schoolers (HS)	-0.36	0.20	0.85	0.92
Adults (AD)	-2.14	0.70	1.11	1.01
Combined Sample (MS, HS, & AD)	-0.52	0.15	0.68**	0.87
Gender DIF (Male) for Combined Sample	0.44	0.19	0.60	0.84
Language DIF (English) for Combined Sample	-0.33	0.18	0.66	0.88
Race DIF (White) for Combined Sample	-0.07	0.13	0.62	0.82
Race DIF (Black) for Combined Sample	-0.07	0.22	0.92	0.98
Race DIF (Other) for Combined Sample	0.14	0.26	0.73	0.94
MotherEd DIF (below HS) for Combined Sample	0.59	0.25	0.77	0.84
MotherEd DIF (HS) for Combined Sample	0.10	0.18	0.89	0.97
MotherEd DIF (College) for Combined Sample	-0.48	0.16	0.60	0.87
MotherEd DIF (Graduate) for Combined Sample	-0.20	0.34	0.37	0.70

13. *Newton – Facet IV – Generating Explanations / Drawing Conclusions

- *Reasoning Skills*

  Learners must understand the theory or law well enough to draw inferences that account for stochastic relationships.

  Task Analysis:

  1. Understand the theory
  2. Understand the comparison/experimental context
  3. Construct (select) an argument that is aligned with the theory and accounts for the experimental outcomes or observed phenomena

- *Description*

  One of Newton's laws suggests that a force (a push or pull) is equal to the mass of the object applying the force multiplied by the acceleration (how the speed of the object is changing every time interval). This law is expressed by: Force=Mass × Acceleration.

a. Andy throws a baseball and a softball on playground. According to Newton's laws, which statement below is true? (Baseball=140g; Softball=200g)

A. If Andy throws a baseball and a softball horizontally with the same force, a softball would have more acceleration than a baseball.
B. If Andy throws a baseball and a softball and he wants to make sure two balls have the same acceleration, he needs to use more force on softball.
C. If Andy throws a baseball twice and he uses more force at the first time than the second time, the ball he throws at the second time will have more acceleration.

b. Another of Newton's laws states that for every action there is an equal and opposite reaction. How could you use this law, to explain the phenomenon that two very large football players pushed against each other but neither moved?

- *Rubric*

a.

Score	Response
1	B
0	A, or C

b.

Score	Response
2	Two players pushed against each other using the same force (strength). That is, the product of the mass and acceleration were the same for them.
1	Their mass (or acceleration) was equal.
0	Wrong answer e.g., they pushed against each other at the same speed.

- *IRT Item Statistics*

a.

	Estimate	SE	Outfit	Infit

Middle Schoolers (MS)	0.89	0.18	1.21	1.16**
High Schoolers (HS)	0.62	0.13	1.24**	1.19**
Adults (AD)	1.57	0.12	1.01	1.00
Combined Sample (MS, HS, & AD)	0.99	0.08	1.24**	1.15**
Gender DIF (Male) for Combined Sample	-0.23	0.09	1.16	1.09
Language DIF (English) for Combined Sample	0.21	0.11	1.15	1.12
Race DIF (White) for Combined Sample	0.19	0.08	1.12	1.09
Race DIF (Black) for Combined Sample	-0.61	0.14	1.11	1.10
Race DIF (Other) for Combined Sample	0.42	0.16	1.32	1.19
MotherEd DIF (below HS) for Combined Sample	0.16	0.14	1.80	1.41
MotherEd DIF (HS) for Combined Sample	0.00	0.11	1.19	1.13
MotherEd DIF (College) for Combined Sample	-0.15	0.09	1.18	1.13
MotherEd DIF (Graduate) for Combined Sample	-0.01	0.20	1.17	1.12

b.

	Estimate	SE	Outfit	Infit
Middle Schoolers (MS)	0.59	0.12	1.47**	1.18
High Schoolers (HS)	0.85	0.09	1.61**	1.30**
Adults (AD)	2.00	0.09	1.06	1.03
Combined Sample (MS, HS, & AD)	1.20	0.05	1.93**	1.37**
Gender DIF (Male) for Combined Sample	-0.07	0.06	2.32	1.41
Language DIF (English) for Combined Sample	0.48	0.07	1.61	1.28
Race DIF (White) for Combined Sample	0.41	0.05	1.66	1.33
Race DIF (Black) for Combined Sample	-0.43	0.09	2.52	1.53
Race DIF (Other) for Combined Sample	0.03	0.10	1.29	1.23
MotherEd DIF (below HS) for Combined Sample	-0.71	0.09	0.86	0.98
MotherEd DIF (HS) for Combined Sample	-0.06	0.07	1.97	1.36
MotherEd DIF (College) for Combined Sample	0.26	0.05	1.66	1.30
MotherEd DIF (Graduate) for Combined Sample	0.52	0.13	1.57	1.29

14. *Organisms – Facet V – Coordinating Theory & Evidence

- *Reasoning Skills*

  In this item, learners must evaluate their theories in light of new observations.

  Task Analysis:

  1. Evaluate initial theory
  2. Recommend changes to theory so as to reconcile theory and evidence

- *Description*

Joan has a theory that all living organisms need oxygen to help produce the energy they need to live. She just read about some new organisms growing in the very deep ocean where there is no oxygen that use sulfur to produce the energy they need to live. How should she revise her theory?

A. Joan does not have to revise it, because organisms in the deep ocean do not matter.

B. Joan's theory could be that all living organisms need oxygen or sulfur to help produce the energy they need to live.

C. Joan's theory could be that all living organisms need sulfur to produce energy to live.

- *Rubric*

Score	Response
1	B
0	A or C

- *IRT Item Statistics*

	Estimate	SE	Outfit	Infit
Middle Schoolers (MS)	-1.56	0.25	0.82	0.97
High Schoolers (HS)	-1.19	0.17	0.87	0.96
Adults (AD)	-1.21	0.32	0.88	1.00
Combined Sample (MS, HS, & AD)	-1.31	0.13	0.77**	0.97
Gender DIF (Male) for Combined Sample	0.21	0.16	0.74	0.97
Language DIF (English) for Combined Sample	0.07	0.15	0.72	0.94
Race DIF (White) for Combined Sample	0.05	0.13	0.80	0.95
Race DIF (Black) for Combined Sample	0.02	0.20	0.80	0.94
Race DIF (Other) for Combined Sample	-0.07	0.23	0.76	0.97
MotherEd DIF (below HS) for Combined Sample	0.34	0.24	1.32	1.10
MotherEd DIF (HS) for Combined Sample	-0.09	0.17	0.87	0.94
MotherEd DIF (College) for Combined Sample	0.00	0.14	0.58	0.91
MotherEd DIF (Graduate) for Combined Sample	-0.25	0.33	0.94	1.12

15. Owls & Rabbits – Facet I – Hypothesis Generation

- *Reasoning Skills*

Learners must show the ability to observe the situation and clearly articulate a question that can direct an empirical investigation

Task Analysis:

1. Note features in observations

2. Make a statement of a possible investigation based on observations

- *Description*

  Below is a drawing of an owl and a rabbit. Owls hunt for rodents and other small animals. Rabbits eat grasses and are often prey for other animals.



  Which of the following would be a reasonable hypothesis about the behavior of owls and rabbits based on their facial characteristics?

  A. Because both the owl and the rabbit have mouths, they eat the same sort of food.
  B. Because the rabbit has whiskers, it can smell better than owls.
  C. Because the owl has bigger ears than the rabbit, it can hear better.
  D. Because the owl has bigger eyes than the rabbit, it can see better in the dark.

- *Rubric*

Score	Response
1	D
0	A, B, or C

- *IRT Item Statistics*

	Estimate	SE	Outfit	Infit
Middle Schoolers (MS)	-0.95	0.19	0.87	0.92
High Schoolers (HS)	-0.72	0.14	0.96	0.97
Adults (AD)	0.05	0.17	0.96	1.00
Combined Sample (MS, HS, & AD)	-0.61	0.10	1.02	1.00
Gender DIF (Male) for Combined Sample	0.00	0.11	1.01	1.01
Language DIF (English) for Combined Sample	-0.04	0.12	1.08	1.01
Race DIF (White) for Combined Sample	0.15	0.10	1.13	1.04
Race DIF (Black) for Combined Sample	-0.29	0.16	0.93	0.92

Race DIF (Other) for Combined Sample	0.14	0.19	0.98	1.04
MotherEd DIF (below HS) for Combined Sample	0.03	0.20	1.18	1.06
MotherEd DIF (HS) for Combined Sample	-0.16	0.14	1.06	1.02
MotherEd DIF (College) for Combined Sample	0.16	0.11	1.11	1.03
MotherEd DIF (Graduate) for Combined Sample	-0.03	0.27	0.78	0.94

16. Parachutes – Facet III – Reasoning from Evidence to Conclusions

- *Reasoning Skills*

  Learners must recognize patterns in the data

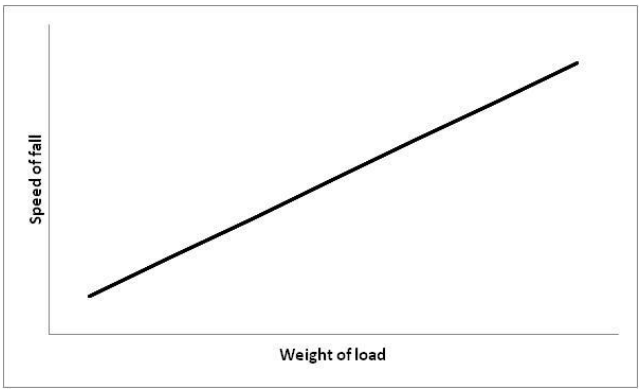
  Task Analysis:

  1. Recognize patterns in the data

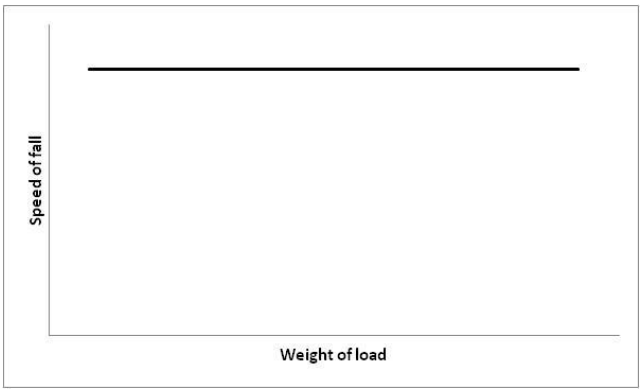
- *Description*

  "The heavier the load, the faster the parachute falls." Which of the graphs below shows this relationship? (*Circle the best answer*)

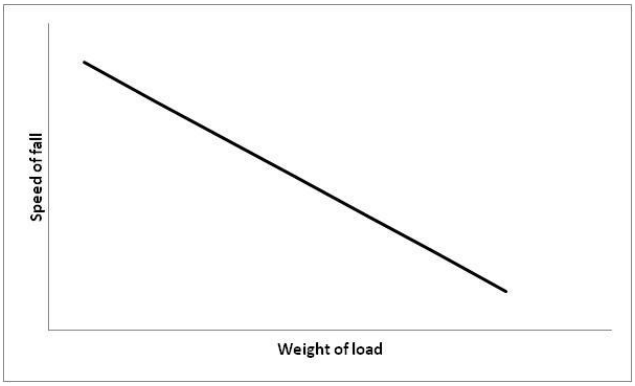
A



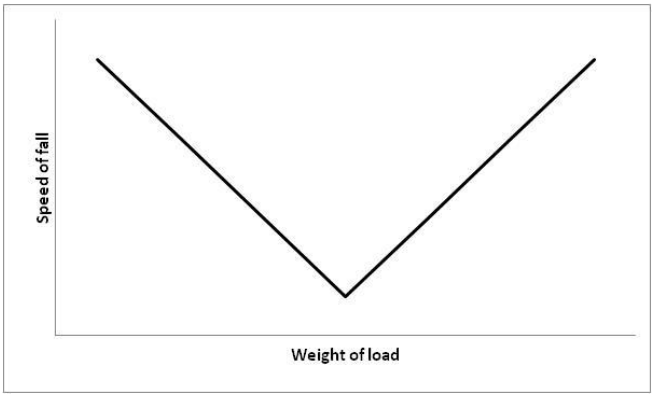
C



B



D



- *Rubric*

Score	Response
1	A
0	B, C or D

- *IRT Item Statistics*

	Estimate	SE	Outfit	Infit
Middle Schoolers (MS)	1.55	0.43	1.25	1.16
High Schoolers (HS)	0.22	0.28	0.96	0.96
Adults (AD)	0.14	0.35	0.99	1.00
Combined Sample (MS, HS, & AD)	0.39	0.18	0.89	0.94
Gender DIF (Male) for Combined Sample	-0.26	0.21	0.73	0.89
Language DIF (English) for Combined Sample	0.06	0.23	0.89	0.93
Race DIF (White) for Combined Sample	0.10	0.14	0.96	0.98
Race DIF (Black) for Combined Sample	-0.20	0.24	0.90	0.95
Race DIF (Other) for Combined Sample	0.10	0.28	0.81	0.78
MotherEd DIF (below HS) for Combined Sample	1.25	0.27	0.71	0.94
MotherEd DIF (HS) for Combined Sample	0.06	0.19	0.90	0.91
MotherEd DIF (College) for Combined Sample	-0.64	0.16	1.03	1.04
MotherEd DIF (Graduate) for Combined Sample	-0.67	0.37	0.76	0.75

17. *Plant Growth – Facet II – Hypothesis Testing

- *Reasoning Skills*

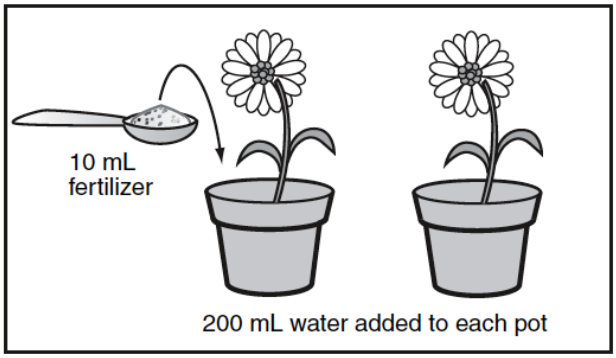
  Learners must observe the experimental situation and formulate a question that can direct an investigation. In this case, the learner is selecting a question that is being investigated.

  Task Analysis:

  1. Observe the experimental situation
  2. Identify factors that vary
  3. Select question that is directing the investigation

- *Description*

  A student will measure and record the growth of two flowering plants every other day for 10 days. According to the diagram, which question is being tested?



A. Do flowering plants grow better when watered with salt water?
B. How much fertilizer do flowering plants need?
C. Does fertilizer added to the soil lead to taller flowering plants?
D. How tall do flowering plants grow?

- *Rubric*

Score	Response
1	C
0	A, B, or D.

- *IRT Item Statistics*

	Estimate	SE	Outfit	Infit
Middle Schoolers (MS)	-1.16	0.49	0.74	0.89
High Schoolers (HS)	-1.18	0.33	0.81	0.90
Adults (AD)	.	.	.	.
Combined Sample (MS, HS, & AD)	-1.35	0.26	0.46**	0.82
Gender DIF (Male) for Combined Sample	0.29	0.29	0.47	0.81
Language DIF (English) for Combined Sample	0.01	0.31	0.46	0.86
Race DIF (White) for Combined Sample	-0.29	0.16	0.47	0.87
Race DIF (Black) for Combined Sample	0.44	0.28	0.84	0.86
Race DIF (Other) for Combined Sample	-0.15	0.32	0.36	0.68
MotherEd DIF (below HS) for Combined Sample	1.24	0.29	1.14	1.08
MotherEd DIF (HS) for Combined Sample	-0.07	0.21	0.59	0.84
MotherEd DIF (College) for Combined Sample	-0.98	0.19	0.35	0.80
MotherEd DIF (Graduate) for Combined Sample	-0.19	0.41	0.24	0.67

18. *Plant Growth Chart – Facet III – Reasoning from Evidence to Conclusions

- *Reasoning Skills*

  Learners must interpret the results of an investigation and draw a justified conclusion based on that data.

  Task Analysis:

  1. Understand the experiment design

  2. Summarize patterns in data table

  3. Draw a justified conclusion / Induce a general statement about the relationships among data which summarizes that evidence

- *Description*

Plant Group	Amount of Light per Day (hours)	Average Growth in One Week (centimeters)
1	4	1
2	6	4
3	8	6
4	10	3

The chart above shows data collected when the same types of potted plants were placed in sunlight for different lengths of time. All other variables, such as the type of plant, size of the pot, soil type, amount of water and fertilizer were kept constant.

Which of the following would be a reasonable claim based on the evidence?

A. There is best amount of light per day in terms of the average growth of plants
B. The amount of light does not affect growth
C. The more sunlight a plant receives, the greater the rate of plant growth.
D. The more sunlight a plant receives, the slower its rate of growth.

- *Rubric*

Score	Response
1	A
0	B, C, or D.

- *IRT Item Statistics*

	Estimate	SE	Outfit	Infit
Middle Schoolers (MS)	0.15	0.30	1.10	1.10
High Schoolers (HS)	0.58	0.28	1.05	1.05

Adults (AD)	0.72	0.28	1.07	1.02
Combined Sample (MS, HS, & AD)	0.46	0.16	1.11	1.05
Gender DIF (Male) for Combined Sample	0.06	0.19	0.97	0.96
Language DIF (English) for Combined Sample	0.13	0.21	1.12	1.03
Race DIF (White) for Combined Sample	-0.05	0.14	1.07	1.02
Race DIF (Black) for Combined Sample	0.27	0.24	1.58	1.37
Race DIF (Other) for Combined Sample	-0.22	0.28	0.98	1.00
MotherEd DIF (below HS) for Combined Sample	0.04	0.27	0.65	0.78
MotherEd DIF (HS) for Combined Sample	-0.08	0.19	1.36	1.17
MotherEd DIF (College) for Combined Sample	0.47	0.15	1.06	1.03
MotherEd DIF (Graduate) for Combined Sample	-0.42	0.36	1.12	0.98

19. *Roy's Lifeboats – Facet II – Hypothesis Testing

- *Reasoning Skills*
  Learners must exhibit the ability to correctly identify and manipulate all relevant variables in order that empirical evidence may be produced that will allow one to make predictions and answer questions.

- *Task Analysis*

  1. Understand research question
  2. Identify what data need to be gathered
  3. Recognize possible confounds
  4. Control variables to determine effect of independent variable
  5. Determine how much measurement is needed
  6. Assess quality of data against what is needed

- *Description*

  Roy wants to find out if a toy lifeboat floating in a basin of salt water can hold more passengers (pennies) than a lifeboat floating in a basin of fresh water. He wrote a materials list and the first step of his investigation, but he is not sure what to do next.

  Roy uses the following steps to find out if having salt in the water affects the number of passengers a lifeboat can hold.

  1. Get two basins of water
  2. Fill one with salt water and one with fresh water
  3. Put a toy lifeboat in the fresh water and carefully add pennies into the boat one at a time until it sinks
  4. Count and record the number of pennies that sank the boat

5. Put the lifeboat in the salt water and carefully add pennies into the boat one at a time until it sinks
6. Count and record the number of pennies that sank the boat
7. Compare the number of pennies it takes to sink the lifeboat in fresh water to the number it takes to sink the lifeboat in salt water

What is incorrect about Roy's steps? Choose one best answer.

A. He should put the pennies in two at a time
B. He should use different lifeboats made from different materials in the salt and regular water
C. He should repeat the fresh water and salt water trials and average the results for each
D. He should put a toy lifeboat in the salt water first.

- *Rubric*

Score	Response
1	C
0	A, B, or D

- *IRT Item Statistics*

	Estimate	SE	Outfit	Infit
Middle Schoolers (MS)	-0.73	0.45	0.69	0.79
High Schoolers (HS)	-1.20	0.34	0.66**	0.81
Adults (AD)	-0.47	0.46	1.03	1.02
Combined Sample (MS, HS, & AD)	-0.94	0.23	0.79**	0.87
Gender DIF (Male) for Combined Sample	0.13	0.27	0.93	0.88
Language DIF (English) for Combined Sample	-0.21	0.27	1.02	0.92
Race DIF (White) for Combined Sample	0.06	0.16	1.09	0.94
Race DIF (Black) for Combined Sample	-0.29	0.28	1.14	0.98
Race DIF (Other) for Combined Sample	0.23	0.32	0.72	0.77
MotherEd DIF (below HS) for Combined Sample	0.46	0.29	0.77	0.86
MotherEd DIF (HS) for Combined Sample	-0.73	0.22	1.18	1.07
MotherEd DIF (College) for Combined Sample	-0.08	0.18	1.09	0.83
MotherEd DIF (Graduate) for Combined Sample	0.35	0.40	1.11	0.88

20. Sugar Water – Facet II – Hypothesis Testing

- *Reasoning Skills*

Learners must design a test of a hypothesis that correctly manipulates the relevant variables in order that empirical evidence may be produced to answer the question.

Task Analysis:

1. Understand experiment design

2. Use CVS to determine effect on dependent variable

3. Suggest changes to create a fair experiment

- *Description*

  Jason wants to know if sugar water freezes faster than plain water. He set up his experiment as follows.

Cup 1	Cup 2
Plastic cup 25 mL water	Paper cup 50 mL water 10 g sugar

  Then Jason put both cups into the freezer at the same time. What changes should Jason make in order to make this a fair experiment? (You may select more than one answer)

  A. The two cups should be the same

  B. The amount of sugar should be the same

  C. The amount of water should be the same

  D. The two cups should be put into the two different freezers at the same time.

- *Rubric*

Score	Response
2	A and C
1	Either A or C
0	Include B or D

- *IRT Item Statistics*

	Estimate	SE	Outfit	Infit
Middle Schoolers (MS)	0.21	0.21	1.23	1.18
High Schoolers (HS)	-0.17	0.22	0.97	1.01
Adults (AD)	0.94	0.30	0.98	0.98
Combined Sample (MS, HS, & AD)	0.31	0.13	1.16	1.15
Gender DIF (Male) for Combined Sample	0.25	0.18	1.09	1.08
Language DIF (English) for Combined Sample	0.03	0.17	1.04	1.04
Race DIF (White) for Combined Sample	0.53	0.11	1.06	1.05

Race DIF (Black) for Combined Sample	-0.81	0.21	1.07	1.07
Race DIF (Other) for Combined Sample	0.28	0.24	0.92	0.96
MotherEd DIF (below HS) for Combined Sample	0.09	0.22	0.56	0.69
MotherEd DIF (HS) for Combined Sample	-0.23	0.16	0.73	0.75
MotherEd DIF (College) for Combined Sample	0.21	0.13	1.16	1.18
MotherEd DIF (Graduate) for Combined Sample	-0.07	0.30	1.23	1.21

21. Tatum's Cakes – Facet II – Hypothesis Testing

- *Reasoning Skills*

  Learners must design a valid test of a hypothesis by recognizing possible confounds

  Task Analysis:

  1. Understand what data need to be gathered

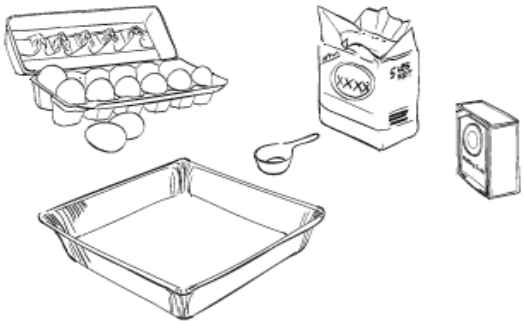
  2. Recognize possible confounds

  3. Make selection that will allow them to control/combine variables to determine the effects on the dependent variable

- *Description*

Tatum likes to bake cakes. Here is the standard cake recipe she uses.

1 teaspoon baking soda
2 eggs
2 cups flour
square pan



Tatum wondered if the amount of baking soda affects how high her cakes rise. Which of the following recipes should she use to find out if baking soda affects how high cakes rise?

(Circle the one best answer.)

A. 2 teaspoons baking soda
2 eggs
2 cups flour
round pan

B. 1 teaspoon baking soda
2 eggs
2 cups flour
round pan

C. 2 teaspoons baking soda
2 eggs
2 cups flour
square pan

D. 2 teaspoons baking soda
2 eggs
1 cup flour
square pan

- *Rubric*

Score	Response
1	C
0	A, B, or D

- *IRT Item Statistics*

	Estimate	SE	Outfit	Infit
Middle Schoolers (MS)	-0.39	0.31	0.93	0.95
High Schoolers (HS)	-0.34	0.30	0.79	0.85
Adults (AD)	-0.74	0.51	0.85	1.00
Combined Sample (MS, HS, & AD)	-0.42	0.19	0.76**	0.87
Gender DIF (Male) for Combined Sample	0.16	0.23	0.81	0.91
Language DIF (English) for Combined Sample	-0.02	0.23	0.80	0.88
Race DIF (White) for Combined Sample	-0.24	0.15	0.88	0.90
Race DIF (Black) for Combined Sample	0.57	0.25	0.75	0.81

Race DIF (Other) for Combined Sample	-0.33	0.30	0.71	1.01
MotherEd DIF (below HS) for Combined Sample	1.63	0.30	0.40	0.47
MotherEd DIF (HS) for Combined Sample	-0.28	0.21	0.57	0.72
MotherEd DIF (College) for Combined Sample	-0.12	0.17	0.89	0.96
MotherEd DIF (Graduate) for Combined Sample	-1.23	0.40	0.58	1.06

22. Taylor's Seedlings – Facet I – Hypothesis Generation

- *Reasoning Skills*

  Learners must observe the experimental situation, and recognize what can be found out by conducting the experiment by noting features and patterns, and identifying factors that vary. Then they can formulate a question that can direct the investigation.

  Task Analysis:

  1. Observe the experimental situation
  2. Note features and patterns
  3. Identify factors that vary and relate factors in a qualitative way
  4. Formulate questions that can direct investigation

- *Description*

  Taylor conducted an experiment using some similar seedlings over a period of two weeks. The table below shows details of his experiment.

Pot	Type of soil	Amount of water used daily (cm^3)	Number of seedlings planted in each pot	Average height of the seedlings in each pot after 2 weeks (cm)
S	Clay soil	100	15	4
T	Potting soil	100	23	8.5
U	Clay soil	100	23	6.5
V	Potting soil	100	15	5

Why do you think that Taylor conducted this experiment? (You may select more than one answer)

A : To find out if overcrowding affects the average height of the seedlings.

B : To find out if different types of soil affect the average height of the seedlings.

C : To find out if different amounts of water affect the average height of the seedlings.

- *Rubric*

Score	Response
2	A and B
1	A or B
0	C, A and C, B and C, or A, B, and C

- *IRT Item Statistics*

	Estimate	SE	Outfit	Infit
Middle Schoolers (MS)	0.11	0.12	1.18	1.13
High Schoolers (HS)	0.16	0.10	1.02	1.03
Adults (AD)	0.88	0.17	0.99	0.99
Combined Sample (MS, HS, & AD)	0.41	0.07	1.13**	1.12**
Gender DIF (Male) for Combined Sample	-0.05	0.08	1.10	1.10
Language DIF (English) for Combined Sample	0.08	0.08	1.09	1.08
Race DIF (White) for Combined Sample	0.25	0.06	1.06	1.05
Race DIF (Black) for Combined Sample	-0.44	0.11	1.01	1.03
Race DIF (Other) for Combined Sample	0.19	0.12	1.06	1.06
MotherEd DIF (below HS) for Combined Sample	0.10	0.11	0.89	0.93
MotherEd DIF (HS) for Combined Sample	-0.28	0.09	1.21	1.17
MotherEd DIF (College) for Combined Sample	-0.07	0.07	1.07	1.08
MotherEd DIF (Graduate) for Combined Sample	0.25	0.16	0.97	0.96

23. *Vitamin C vs. B12 – Facet V – Coordinating Theory & Evidence

- *Reasoning Skills*

Learners must evaluate a theory in light of an experimental conclusion, reconcile new evidence with prior beliefs and (if required) revise one's theory and generate new predictions.

Task Analysis

1. Interpret the experiment
2. Reflect on strengths and weaknesses of the theory and evidence (experimental outcomes)

3. Assess theory and evidence for correlational relationship

- *Description*

Martina has a theory that Vitamin C helps to cure common colds. She tests her theory by studying people with colds. She randomly selects 150 people who have colds. Then she divides the group of 150 into three different groups. The first group of 50 people take Vitamin C every day until their colds go away. The second group of 50 people take Vitamin B12 every day until their colds go away. The third group of 50 people takes neither Vitamin B12 nor Vitamin C until their colds go away.

	Ave number of days with a cold
Vitamin C people	13 days
B12 people	14 days
No vitamin people	13 days

Which of the following would you pick as a statement of the theory based on the evidence?

A. The evidence shows that Vitamin C helps to cure common colds

B. The evidence shows that Vitamin C may help to cure common colds better than Vitamin B12

C. The evidence shows that Vitamin C has no different effect on curing common colds than Vitamin B12 or no vitamins.

- *Rubric*

Score	Response
1	C
0	A or B

- *IRT Item Statistics*

	Estimate	SE	Outfit	Infit
Middle Schoolers (MS)	1.43	0.25	1.25	1.10
High Schoolers (HS)	0.95	0.16	1.41**	1.31**
Adults (AD)	2.05	0.14	1.03	1.02
Combined Sample (MS, HS, & AD)	1.44	0.10	1.36**	1.22**
Gender DIF (Male) for Combined Sample	-0.17	0.11	1.38	1.19
Language DIF (English) for Combined Sample	0.58	0.14	1.27	1.18
Race DIF (White) for Combined Sample	0.54	0.09	1.30	1.18
Race DIF (Black) for Combined Sample	-0.62	0.17	1.28	1.24
Race DIF (Other) for Combined Sample	0.07	0.19	1.16	1.08
MotherEd DIF (below HS) for Combined Sample	-0.46	0.17	1.87	1.56

MotherEd DIF (HS) for Combined Sample	-0.14	0.13	1.23	1.18
MotherEd DIF (College) for Combined Sample	0.22	0.10	1.42	1.20
MotherEd DIF (Graduate) for Combined Sample	0.38	0.23	1.03	1.04

24. Water Cycle – Facet IV – Generating Explanations / Drawing Conclusions

- *Reasoning Skills*

Learners must describe or explain the underlying cause for an observed phenomenon.

Task Analysis:

1. Understand the experiment context
2. Recognize that the "data" or observation is in conflict with the expected observation
3. Apply a scientific principle to account for the observed phenomenon

Jane went for a walk on a 90 degree day with high humidity and got very wet and sweaty. She went for a walk in another city on a 90 degree day with low humidity and she did not get wet and sweaty. People sweat when they exercise so that the evaporation of the sweat can cool them off.

Jane's clothes got wet when she went walking on the high-humidity day but did not get wet when she went walking on the low-humidity day. Which of the following is the best explanation?

A. Because the high humidity air already contains a lot of water vapor and cannot hold much more water, the sweat cannot evaporate.
B. Your body does not produce sweat on low humidity days
C. Because high humidity air has more water vapor in the air, it condensed onto Jane making her wet.
D. Because the high humidity air has less water vapor, Jane's sweat would evaporate

- *Rubric*

Score	Response
1	A
0	B, C, or D

- *IRT Item Statistics*

	Estimate	SE	Outfit	Infit
Middle Schoolers (MS)	0.79	0.40	1.08	1.09
High Schoolers (HS)	1.52	0.31	0.99	1.01
Adults (AD)	1.30	0.24	1.02	1.01
Combined Sample (MS, HS, & AD)	1.12	0.17	1.02	1.00
Gender DIF (Male) for Combined Sample	0.12	0.18	1.11	1.08
Language DIF (English) for Combined Sample	-0.33	0.27	1.12	1.06

Race DIF (White) for Combined Sample	-0.53	0.13	1.09	1.04
Race DIF (Black) for Combined Sample	0.94	0.26	2.15	1.17
Race DIF (Other) for Combined Sample	-0.41	0.29	1.18	1.11
MotherEd DIF (below HS) for Combined Sample	0.54	0.25	0.82	0.97
MotherEd DIF (HS) for Combined Sample	-0.46	0.19	1.23	1.18
MotherEd DIF (College) for Combined Sample	-0.04	0.15	1.12	1.06
MotherEd DIF (Graduate) for Combined Sample	-0.04	0.34	0.94	0.98

25. Sam's Plants – Facet II – Hypothesis Testing

- *Reasoning Skills*

  Learners must identify a test that correctly identifies and manipulates all relevant variables in order that evidence may be produced that will allow one to answer questions.

  Task Analysis:

  1. Identify what data need to be gathered

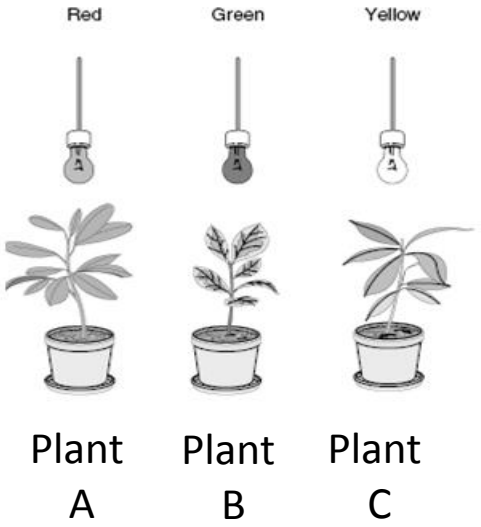
  2. Recognize possible confounds

  3. Control variables to determine an effect on the dependent variable

  4. Suggest best way to improve the experiment

- *Description*

  Sam conducted an experiment to test the hypothesis that plants grow faster in green light. The same amount of water and fertilizer were added to each pot. He set up the experiment as shown below and found out that the Plant A grew faster than the other two. Therefore, he concluded that the hypothesis is wrong and plants grow faster in red light but he is not quite sure. He decides to change the experiment and try again.



What is the best way to improve the experiment?

A. Use only red and green light

B. Use only plant A with red, green and yellow light.

C. Use plant A and plant B with red and green light.

D. Use different amounts of water and fertilizer.

- *Rubric*

Score	Response
1	B
0	A, C, or D

- *IRT Item Statistics*

	Estimate	SE	Outfit	Infit
Middle Schoolers (MS)	0.77	0.17	1.14	1.08
High Schoolers (HS)	0.71	0.13	1.05	1.04
Adults (AD)	0.72	0.14	1.00	1.00
Combined Sample (MS, HS, & AD)	0.62	0.08	0.98	0.98
Gender DIF (Male) for Combined Sample	0.21	0.09	1.08	1.04
Language DIF (English) for Combined Sample	-0.22	0.11	1.01	1.00
Race DIF (White) for Combined Sample	0.33	0.08	0.96	0.95
Race DIF (Black) for Combined Sample	-0.28	0.14	1.22	1.18
Race DIF (Other) for Combined Sample	-0.05	0.16	1.12	1.06
MotherEd DIF (below HS) for Combined Sample	0.56	0.15	0.93	0.93
MotherEd DIF (HS) for Combined Sample	-0.53	0.11	1.05	1.11
MotherEd DIF (College) for Combined Sample	-0.26	0.09	0.94	0.97
MotherEd DIF (Graduate) for Combined Sample	0.22	0.21	1.12	1.01

26. *Baking a Cake – Facet II – Hypothesis Testing

- *Reasoning Skills*

Learners must identify a test that correctly identifies and manipulates all relevant variables in order that empirical evidence may be produced that will allow one to answer question

Task Analysis:

1. Identify what data need to be gathered
2. Recognize possible confounds
3. Control variables to determine an effect on the dependent variable

- *Description*

John decided to bake a cake. The cake recipe called for butter for the shortening, sugar for the sweetening, and white flour. John decided to make a change to the recipe.

- He used margarine instead of butter for the shortening
- He used honey instead of sugar for the sweetening, and
- He used brown whole wheat flour instead of regular white flour

The cake turned out great. It was so moist. John thought that the reason the cake was so great was the honey. He thought that the type of shortening (butter or margarine) or the type of flour really didn't matter. What should he do to prove this point? (*Circle the best answer*)

A. Bake the cake again but use sugar instead of honey, and still use the margarine and brown whole wheat flour.

B. Bake the cake again but this time use sugar, butter, and regular white flour.

C. Bake the cake again still using honey, but this time using butter and regular white flour.

- *Rubric*

Score	Response
1	A
0	B or C

- *IRT Item Statistics*

	Estimate	SE	Outfit	Infit
Middle Schoolers (MS)	1.68	0.46	1.12	1.12
High Schoolers (HS)	1.58	0.29	1.24	1.16
Adults (AD)	3.49*	0.23	1.03	1.03
Combined Sample (MS, HS, & AD)	2.60*	0.17	1.50**	1.25**
Gender DIF (Male) for Combined Sample	-0.01	0.20	1.63	1.30
Language DIF (English) for Combined Sample	0.92	0.24	1.44	1.22
Race DIF (White) for Combined Sample	1.27	0.13	1.25	1.13
Race DIF (Black) for Combined Sample	-1.41	0.24	1.26	1.24
Race DIF (Other) for Combined Sample	0.14	0.27	1.23	1.14
MotherEd DIF (below HS) for Combined Sample	-1.67	0.25	1.60	1.47
MotherEd DIF (HS) for Combined Sample	0.07	0.19	1.37	1.22
MotherEd DIF (College) for Combined Sample	0.23	0.15	1.21	1.12
MotherEd DIF (Graduate) for Combined Sample	1.37	0.35	1.26	1.14

27. Cavities by Country – Facet III – Reasoning from Evidence to Conclusions

- *Reasoning Skills*

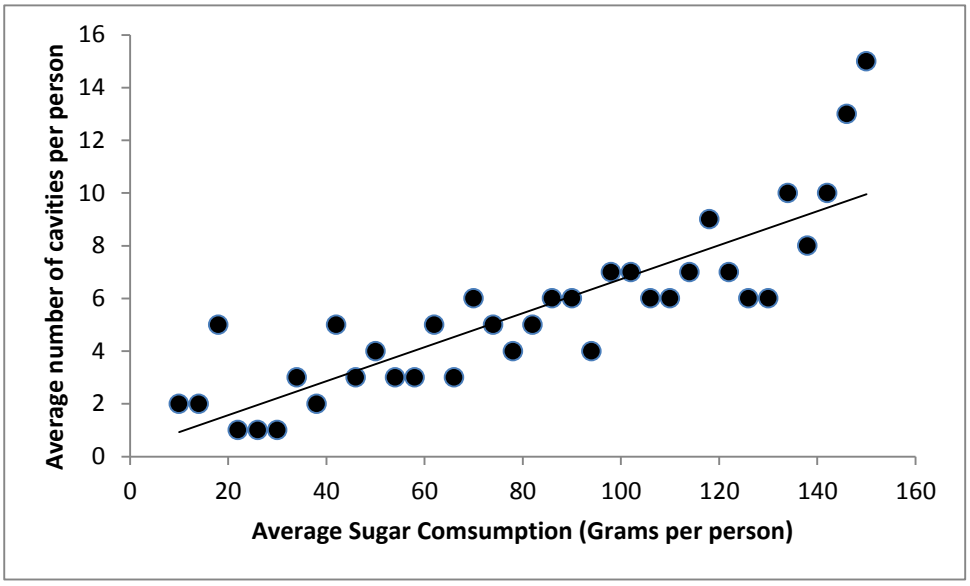
Learners must draw correct conclusions based on a graphical data representation.

Task Analysis:

1. Interpret graph
2. Recognize patterns in the data

- *Description*

The following graph shows the consumption of sugar and the number of cavities (decayed teeth) in different countries. Each country is represented by a dot in the graph.



Which one of the following statements is *best* supported by the data given in the graph? *Choose one answer.*

A. In some countries, people brush their teeth more frequently than in other countries.

B. The more sugar people eat, the more likely they are to get cavities.

C. In recent years, the rate of cavities has increased in many countries.

D. In recent years, the consumption of sugar has increased in many countries.

- *Rubric*

Score	Response
1	B
0	A, C, D

- *IRT Item Statistics*

	Estimate	SE	Outfit	Infit
Middle Schoolers (MS)	-0.36	0.40	0.71	0.79
High Schoolers (HS)	-0.03	0.28	0.98	0.98
Adults (AD)	-2.18	0.99	0.89	1.02

Combined Sample (MS, HS, & AD)	-0.45	0.20	0.60**	0.83
Gender DIF (Male) for Combined Sample	-0.13	0.24	0.57	0.79
Language DIF (English) for Combined Sample	-0.01	0.26	0.55	0.77
Race DIF (White) for Combined Sample	-0.03	0.15	0.49	0.80
Race DIF (Black) for Combined Sample	0.11	0.27	0.71	0.80
Race DIF (Other) for Combined Sample	-0.08	0.31	0.54	0.67
MotherEd DIF (below HS) for Combined Sample	-7.60	0.31	.	.
MotherEd DIF (HS) for Combined Sample	2.91	0.21	0.64	0.74
MotherEd DIF (College) for Combined Sample	2.50	0.17	0.60	0.83
MotherEd DIF (Graduate) for Combined Sample	2.20	0.41	0.27	0.70

28. Pea Seeds – Facet II – Hypothesis Testing

- *Reasoning Skills*

  Learners must design or critique a test of a hypothesis that correctly manipulates the relevant variables.

  Task Analysis:

  1. Understand experimental context

  2. Classify data into dependent, independent, or controls

  3. Assess the quality and variety of data collected against what is needed

- *Description*

  One hundred pea seeds were put in small dishes and covered with wet paper towels. Fifty of them were placed in a cool room. The other fifty of the pea seeds were placed in a warm room.

  a. This experiment was most likely designed to study the effect of which variable on the sprouting of pea seeds? *Choose the best answer.*

  A. temperature

  B. water

  C. light

  D. seed type

  b. Explain your choice.

- *Rubric*

Score	Response
2	A, since the only difference between two conditions is the temperature
1	A, wrong or no explanations
0	B, C, or D

- *IRT Item Statistics*

	Estimate	SE	Outfit	Infit
Middle Schoolers (MS)	-0.54	0.13	0.83	0.92
High Schoolers (HS)	-0.67	0.10	0.80**	0.84
Adults (AD)	-0.33	0.35	1.14**	1.01
Combined Sample (MS, HS, & AD)	-0.56	0.08	0.90**	0.90
Gender DIF (Male) for Combined Sample	0.32	0.10	1.00	0.88
Language DIF (English) for Combined Sample	0.13	0.09	1.05	0.87
Race DIF (White) for Combined Sample	-0.01	0.08	1.21	0.90
Race DIF (Black) for Combined Sample	-0.01	0.11	0.85	0.90
Race DIF (Other) for Combined Sample	0.02	0.14	0.81	0.97
MotherEd DIF (below HS) for Combined Sample	-0.02	0.16	0.92	0.90
MotherEd DIF (HS) for Combined Sample	-0.02	0.11	1.00	0.92
MotherEd DIF (College) for Combined Sample	0.01	0.09	0.95	0.91
MotherEd DIF (Graduate) for Combined Sample	0.03	0.22	1.13	0.77

29. Pendulums – Facet IV – Generating Explanations / Drawing Conclusions

- *Reasoning Skills*

  Learners must control variables to draw a conclusion.

  Task Analysis:

  1. Part a – Identify anomalies and inconsistencies in the explanation
  2. Part c – 2.1a – Identify what data need to be gathered

- *Description*

  Sophia was playing on a swing set at a park. The park had three different swing sets, each one of different length. She was curious about how the length of the swing set affected the number of swings. She kept track of the following information about the three swings.

Swing Set	Length of Swing (feet)	Number of Swings/Minute

A	10 ft	25
B	7 ft	35
C	5 ft	45

a. Which of the following seems to be true?

A. As the swing set gets longer, the number of swings increases.

B. As the swing set gets longer, the number of swings decreases.

C. The longer swing set, then the number of swings can either increase or decrease.

b. If Sophia makes a new swing set that is 15 feet long, how many swings will there be in a minute? *Choose the best answer.*

A. Less than 25

B. More than 25

C. More than 45

D. Between 25 and 45

c. Sophia wants to know if a heavier child on the swing makes a difference in the number of swings. What is the best way to test the effect of the weight of the child on number of swings?

A. Change the length of the swing set.

B. Use a heavier child.

C. Change the length of the swing set and use a heavier child.

- *Rubric*

a.

Score	Response
1	B
0	A or C

b.

Score	Response
1	A
0	B, C, or D

c.

Score	Response
1	B
0	A or C

- *IRT Item Statistics*

a.

	Estimate	SE	Outfit	Infit
Middle Schoolers (MS)	-1.28	0.22	0.91	0.98
High Schoolers (HS)	-1.09	0.16	0.80**	0.91
Adults (AD)	-3.06*	0.69	0.88	1.00
Combined Sample (MS, HS, & AD)	-1.32	0.12	0.61**	0.89
Gender DIF (Male) for Combined Sample	0.04	0.15	0.42	0.83
Language DIF (English) for Combined Sample	-0.25	0.13	0.60	0.92
Race DIF (White) for Combined Sample	-0.03	0.13	0.61	0.93
Race DIF (Black) for Combined Sample	0.41	0.19	0.79	0.89
Race DIF (Other) for Combined Sample	-0.37	0.23	0.53	0.90
MotherEd DIF (below HS) for Combined Sample	0.83	0.25	0.97	0.95
MotherEd DIF (HS) for Combined Sample	0.12	0.17	0.90	0.94
MotherEd DIF (College) for Combined Sample	-0.11	0.14	0.57	0.91
MotherEd DIF (Graduate) for Combined Sample	-0.84	0.33	0.13	0.88

b.

	Estimate	SE	Outfit	Infit
Middle Schoolers (MS)	-1.39	0.22	0.73**	0.87
High Schoolers (HS)	-1.06	0.16	0.84**	0.90
Adults (AD)	-2.65*	0.57	1.06	1.00
Combined Sample (MS, HS, & AD)	-1.33	0.12	0.60**	0.87
Gender DIF (Male) for Combined Sample	0.11	0.15	0.50	0.84
Language DIF (English) for Combined Sample	-0.13	0.14	0.54	0.90
Race DIF (White) for Combined Sample	-0.20	0.12	0.51	0.93
Race DIF (Black) for Combined Sample	0.26	0.18	0.81	0.87
Race DIF (Other) for Combined Sample	-0.06	0.22	0.78	0.90
MotherEd DIF (below HS) for Combined Sample	0.68	0.23	0.52	0.71
MotherEd DIF (HS) for Combined Sample	-0.10	0.16	0.47	0.82
MotherEd DIF (College) for Combined Sample	-0.29	0.14	0.52	0.88
MotherEd DIF (Graduate) for Combined Sample	-0.29	0.31	1.01	1.06

c.

	Estimate	SE	Outfit	Infit
Middle Schoolers (MS)	-0.14	0.18	1.07	1.05

High Schoolers (HS)	-0.29	0.14	0.90	0.94
Adults (AD)	-0.03	0.18	1.00	1.00
Combined Sample (MS, HS, & AD)	-0.24	0.09	0.96	0.97
Gender DIF (Male) for Combined Sample	0.01	0.10	0.81	0.96
Language DIF (English) for Combined Sample	0.03	0.11	0.95	0.96
Race DIF (White) for Combined Sample	-0.08	0.09	0.94	0.97
Race DIF (Black) for Combined Sample	-0.11	0.15	1.02	1.03
Race DIF (Other) for Combined Sample	0.19	0.17	1.09	1.04
MotherEd DIF (below HS) for Combined Sample	0.81	0.19	1.31	1.29
MotherEd DIF (HS) for Combined Sample	-0.08	0.13	0.99	0.96
MotherEd DIF (College) for Combined Sample	-0.30	0.11	0.90	0.94
MotherEd DIF (Graduate) for Combined Sample	-0.43	0.25	1.44	1.18

30. *Wind Farm – Facet III – Reasoning from Evidence to Conclusions

- *Reasoning Skills*

  Learners must have the ability to interpret the results of an investigation and to draw a conclusion.

  Task Analysis:

  1. Recognize patterns in the data
  2. Draw a justified conclusion based on that data

- *Description*

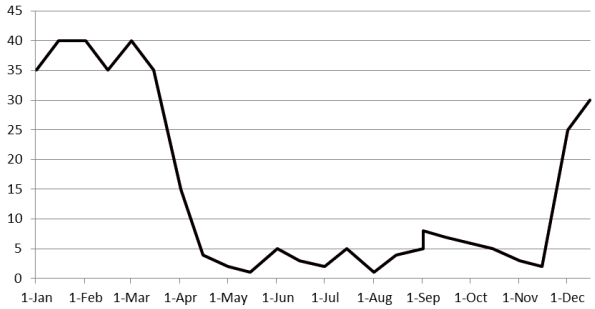
  Many people believe that wind should replace oil and coal as a source of energy for producing electricity. The structures in the picture are windmills with blades that are rotated by the wind. These rotations cause electricity to be produced by generators that are turned by the windmills.



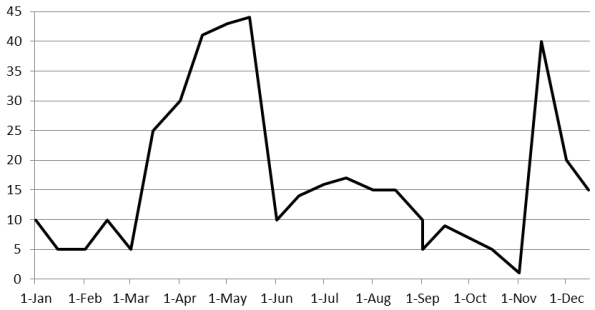
A wind farm

The graphs below show the average wind speeds in four different places throughout a year. Which one of the graphs indicates the most appropriate place to establish a wind farm for generating electricity? *Circle your answer.*

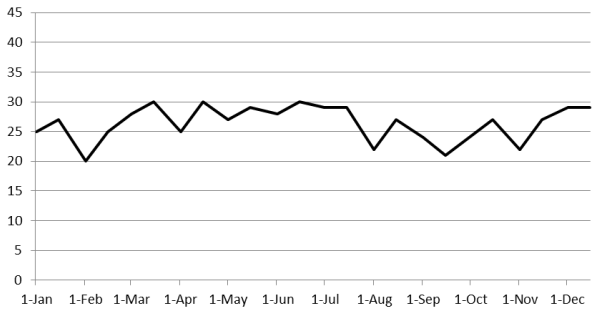
A.



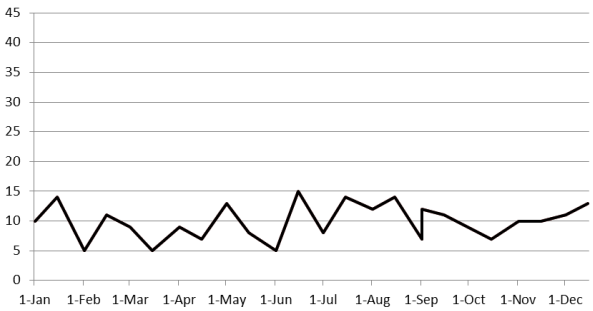
B.



C.



D.



- *Rubric*

Score	Response
1	C
0	A, B, or D

- *IRT Item Statistics*

	Estimate	SE	Outfit	Infit
Middle Schoolers (MS)	-0.32	0.18	0.83	0.88
High Schoolers (HS)	-0.38	0.14	0.90	0.92
Adults (AD)	-1.79	0.38	0.72**	0.99
Combined Sample (MS, HS, & AD)	-0.59	0.10	0.60**	0.83**
Gender DIF (Male) for Combined Sample	0.07	0.12	0.55	0.80
Language DIF (English) for Combined Sample	-0.10	0.12	0.56	0.82
Race DIF (White) for Combined Sample	-0.50	0.11	0.58	0.89
Race DIF (Black) for Combined Sample	0.30	0.15	0.82	0.86
Race DIF (Other) for Combined Sample	0.20	0.18	0.68	0.85
MotherEd DIF (below HS) for Combined Sample	0.40	0.21	0.80	0.93
MotherEd DIF (HS) for Combined Sample	0.01	0.14	0.56	0.78
MotherEd DIF (College) for Combined Sample	-0.10	0.12	0.67	0.88
MotherEd DIF (Graduate) for Combined Sample	-0.31	0.28	0.62	0.86

31. *Wooden slope – Facet II – Hypothesis Testing

- *Reasoning Skills*

  Learners must understand how to design a test of a hypothesis that correctly identifies and manipulates all relevant

  Task Analysis:

  1. Understand task goal
  2. Identify what data need to be gathered
  3. Determine how much measurement is needed for reliability

- *Description*

  A group of students are given a stopwatch and are asked to measure the time that the ball takes from the edge of the table to hitting the ground after being released at h = 400 mm.

  They discuss what to do:

  Student A: "We can roll the ball once from h = 400 mm and measure the time. Once is enough."

  Student B: "Let's roll the ball twice from height h = 400 mm, and measure the time for both rolls."

  Student C: "I think we should release the ball more than twice from h = 400 mm and measure the time."

  Which student do you agree with more? (*Circle one*) A B C

  Explain your choice.

- *Rubric*

Score	Response
2	B or C, they need to make sure that the measure that they get is consistent (or they can get more accurate answer).
1	B or C, incorrect or no explanation A, they will know the answer to the question
0	A, no explanation

- *IRT Item Statistics*

	Estimate	SE	Outfit	Infit
Middle Schoolers (MS)	-1.36	0.39	1.21	1.15
High Schoolers (HS)	-0.69	0.22	0.98	0.98
Adults (AD)	-0.08	0.49	1.00	1.00
Combined Sample (MS, HS, & AD)	-0.78	0.18	1.13	1.07
Gender DIF (Male) for Combined Sample	-0.05	0.21	1.14	1.15
Language DIF (English) for Combined Sample	-0.02	0.21	1.12	1.04
Race DIF (White) for Combined Sample	0.30	0.11	1.02	0.99
Race DIF (Black) for Combined Sample	-0.09	0.19	1.34	1.26
Race DIF (Other) for Combined Sample	-0.20	0.22	1.35	1.14
MotherEd DIF (below HS) for Combined Sample	-0.70	0.21	0.33	0.73
MotherEd DIF (HS) for Combined Sample	-0.20	0.15	0.98	1.07
MotherEd DIF (College) for Combined Sample	0.35	0.12	1.14	1.04
MotherEd DIF (Graduate) for Combined Sample	0.55	0.28	1.22	1.10

32. Catalytic Converter – Facet IV – Generating Explanations / Drawing Conclusions

- *Reasoning Skills*

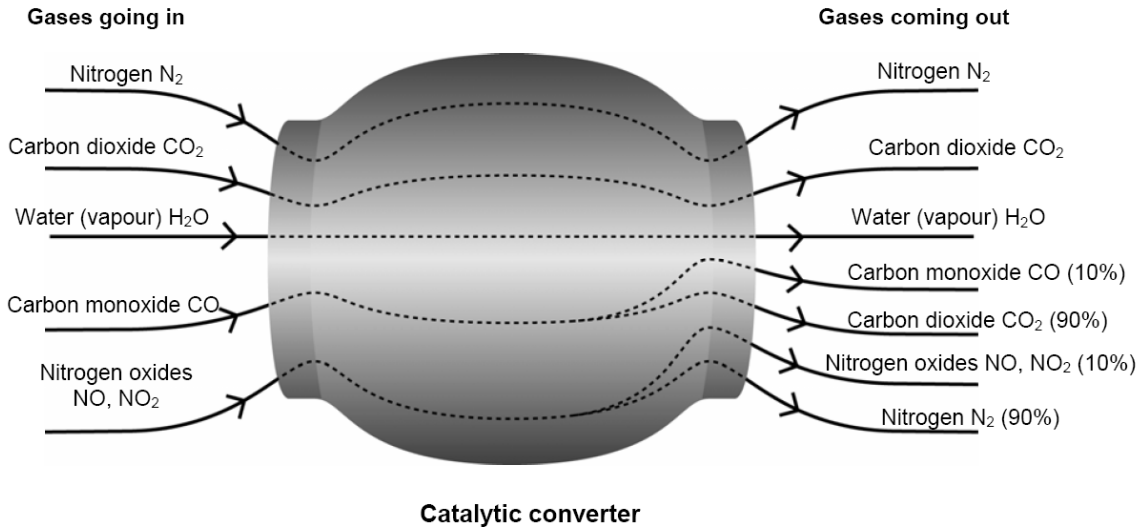
  Learners must coordinate theory and evidence in such a way so as to draw inferences that account for a causal relationship.

  Task Analysis:

  1. Understand theory

  2. Construct (select) an argument that is aligned with the theory and accounts for the experimental outcomes or observed phenomenon

- *Description*



a. Based on the information in the diagram above, select one example of how the catalytic converter makes exhaust fumes *less harmful*.

A. Carbon dioxide is changed into carbon monoxide.
B. Nitrogen oxides are changed into nitrogen.
C. Water vapor is produced
D. The amount of carbon monoxide is increased.

b. Examine the gases emitted by the catalytic converter. What could be done to make the converter produce even less harmful exhaust fumes?

A. Changing carbon dioxide completely into carbon monoxide
B. Producing less water vapor
C. Decreasing the amount of nitrous oxides produced
D. Decreasing the amount of nitrogen produced

- *Rubric*

a.

Score	Response
1	B
0	A, C, or D

b.

Score	Response
1	C
0	A, B, or D

- *IRT Item Statistics*

a.

	Estimate	SE	Outfit	Infit
Middle Schoolers (MS)	1.01	0.31	0.84	0.87
High Schoolers (HS)	0.66	0.30	0.97	0.99
Adults (AD)	0.14	0.35	0.85	0.98
Combined Sample (MS, HS, & AD)	0.56	0.17	0.77**	0.86
Gender DIF (Male) for Combined Sample	-0.03	0.19	0.79	0.86
Language DIF (English) for Combined Sample	-0.59	0.24	0.81	0.91
Race DIF (White) for Combined Sample	-0.25	0.14	0.77	0.92
Race DIF (Black) for Combined Sample	0.88	0.25	0.98	0.97
Race DIF (Other) for Combined Sample	-0.63	0.29	0.47	0.61
MotherEd DIF (below HS) for Combined Sample	-0.12	0.24	0.56	0.70
MotherEd DIF (HS) for Combined Sample	-0.47	0.18	0.58	0.72

MotherEd DIF (College) for Combined Sample	-0.23	0.15	0.74	0.84
MotherEd DIF (Graduate) for Combined Sample	0.81	0.34	0.80	0.89

b.

	Estimate	SE	Outfit	Infit
Middle Schoolers (MS)	0.55	0.30	1.02	1.01
High Schoolers (HS)	1.03	0.31	1.03	1.02
Adults (AD)	0.81	0.28	0.99	1.00
Combined Sample (MS, HS, & AD)	0.73	0.17	1.00	1.00
Gender DIF (Male) for Combined Sample	0.10	0.19	0.97	0.95
Language DIF (English) for Combined Sample	-0.60	0.24	1.08	1.02
Race DIF (White) for Combined Sample	-0.12	0.13	1.09	1.04
Race DIF (Black) for Combined Sample	0.49	0.25	0.87	0.95
Race DIF (Other) for Combined Sample	-0.36	0.28	1.00	1.03
MotherEd DIF (below HS) for Combined Sample	1.65	0.27	0.34	0.54
MotherEd DIF (HS) for Combined Sample	-0.62	0.19	0.94	0.94
MotherEd DIF (College) for Combined Sample	-0.24	0.15	1.03	1.03
MotherEd DIF (Graduate) for Combined Sample	-0.79	0.36	1.00	1.12

33. Emission Reduction – Facet IV – Generating Explanations / Drawing Conclusions

- *Reasoning Skills*

  Learners must have the ability to interpret the results of an investigation and to draw justified inferences/conclusions based upon that data.

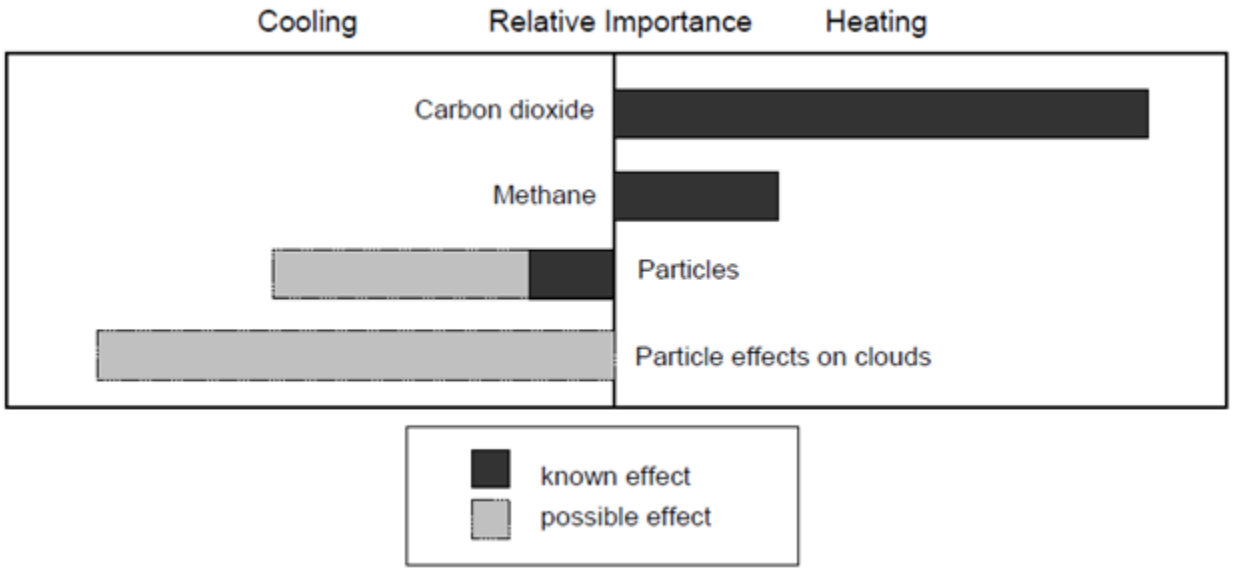
  Task Analysis:

  1. Recognize patterns in the data all that is involved
  2. Interpret data to draw a conclusion.

- *Description*

WHAT HUMAN ACTIVITIES CONTRIBUTE TO CLIMATE CHANGE?

The burning of coal, oil and natural gas, as well as deforestation and various agricultural and industrial practices, are altering the composition of the atmosphere and contributing to climate change. These human activities have led to increased concentrations of particles and greenhouse gases in the atmosphere. The relative importance of the main contributors to temperature change is shown in the Figure below. Increased concentrations of carbon dioxide and methane have a heating effect. Increased concentrations of particles have a cooling effect in two ways, labeled ‘Particles’ and ‘Particle effects on clouds’.



The relative effect of 'Particles' and 'Particle effects on clouds' are quite uncertain: in each case the possible effect is somewhere in the range shown by the light grey bar.

Based on the information in the Figure above, what would be most effective way to reduce the heating effect?

A. Increasing the particle effects on clouds since it has greatest possible cooling effect.
B. Increasing the particles in the atmosphere since it has both known and possible cooling effect.
C. Decreasing the methane in the atmosphere since it has heating effect.
D. Decreasing the carbon dioxide in the atmosphere since it has the greatest heating effect.

- *Rubric*

1	D
0	A, B, or C

- *IRT Item Statistics*

	Estimate	SE	Outfit	Infit
Middle Schoolers (MS)	1.00	0.43	1.53	1.26
High Schoolers (HS)	1.11	0.30	0.99	1.00
Adults (AD)	1.27	0.25	1.00	1.01
Combined Sample (MS, HS, & AD)	0.98	0.17	0.98	1.00
Gender DIF (Male) for Combined Sample	-0.04	0.19	0.97	0.97
Language DIF (English) for Combined Sample	-0.05	0.25	0.96	0.97
Race DIF (White) for Combined Sample	-0.09	0.13	1.04	1.03
Race DIF (Black) for Combined Sample	-0.38	0.26	1.31	1.20
Race DIF (Other) for Combined Sample	0.46	0.29	0.92	0.99
MotherEd DIF (below HS) for Combined Sample	-1.40	0.27	0.31	0.50

MotherEd DIF (HS) for Combined Sample	0.94	0.19	0.85	0.92
MotherEd DIF (College) for Combined Sample	0.26	0.15	1.13	1.09
MotherEd DIF (Graduate) for Combined Sample	0.19	0.36	0.98	0.98

34. *Spring Experiment – Facet III – Reasoning from Evidence to Conclusions

- *Reasoning Skills*

  Learners must have the ability to interpret the results of an investigation and to draw justified inferences/conclusions based upon that data

  Task Analysis:

  1. Recognize patterns in the data

  2. Construct (identify) a scientific argument based upon evidence showing how the evidence supports the conclusion

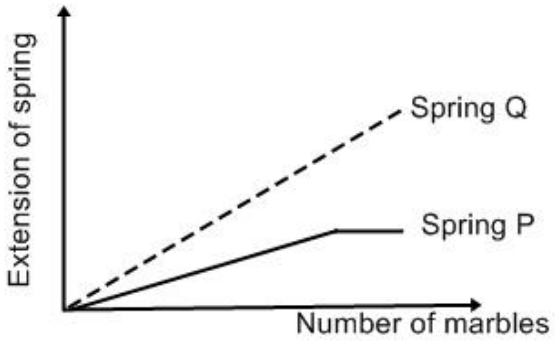
- *Description*

  The diagram below shows an experimental setup using two springs, Spring P and Spring Q. The springs were exactly the same except they were made of different materials.



SPRING P SPRING Q

Similar numbers of marbles were placed in each pan and the length of the spring was recorded. The results of the two springs are shown in the graph below.



Based on the results in the graph, the following statements were made by four pupils.

Alice : Spring Q was stronger than Spring P.
Bala : Spring P extended less than Spring Q for the same number of marbles added.
Christine : Both springs had reached their maximum extended length.
Dollah : Both springs had the same original length.

Who had made the correct statement?

A. Alice only
B. Bala only
C. Alice and Christine only
D. Bala and Dollah only

- *Rubric*

Score	Response
1	D
0	A, B, or C

- *IRT Item Statistics*

	Estimate	SE	Outfit	Infit
Middle Schoolers (MS)	1.51	0.44	1.10	1.06
High Schoolers (HS)	1.44	0.31	1.16	1.11
Adults (AD)	1.12	0.26	1.04	1.02
Combined Sample (MS, HS, & AD)	1.12	0.17	0.95	0.96
Gender DIF (Male) for Combined Sample	-0.16	0.18	1.00	0.95
Language DIF (English) for Combined Sample	0.03	0.24	0.95	0.96
Race DIF (White) for Combined Sample	-0.50	0.79	1.06	1.03
Race DIF (Black) for Combined Sample	1.00	1.37	0.86	1.10
Race DIF (Other) for Combined Sample	-0.50	1.58	1.20	1.09
MotherEd DIF (below HS) for Combined Sample	-0.48	1.52	0.53	0.58
MotherEd DIF (HS) for Combined Sample	0.107*	1.08	0.92	0.92
MotherEd DIF (College) for Combined Sample	0.11	0.89	1.03	0.99
MotherEd DIF (Graduate) for Combined Sample	0.27	2.07	1.21	1.15

Appendix A

Description of IRT Statistics

- Estimate

  The item estimates correspond to the IRT item difficulty with higher estimates being indicative of more difficult items. It usually ranges from -3 to +3, and in this study the absolute value of estimates of 2.5 was used as the cutoff to identify items with an extreme difficulty or easiness (note: * next to the estimate indicates the extreme difficulty or easiness).

- Standard Error (SE)

  SE is an IRT statistic that can be interpreted in the same way as the standard error of measurement (SEM) of Classical Test Theory. The SE is equal to the reciprocal of the square root of the Item Information Function, i.e., $E(\theta) = \frac{1}{\sqrt{I(\theta)}} = \frac{1}{\sqrt{P(\theta)Q(\theta)}}$, where P (θ) is the probability of a correct response of the item for the given θ, and Q (θ) is the probability of an incorrect response of the item for the given θ. Ideally, the standard errors of item and step parameter estimates should be less than the absolute value of estimates divided by 3.5.

- Infit and Outfit

  MNSQ fit statistics, named as infit and outfit, are residual-based indices that are similar in conception and purpose to the weighted and unweighted fit statistics that were developed for Rasch's simple logistic model and the partial credit model. For those models, they can be used to test the violations of the model assumptions of no guessing (infit) and homogeneity of item discrimination (outfit). Outfit is an outlier-sensitive fit statistic; it is sensitive to unexpected

responses that are relatively very easy or very difficult for that person. It can therefore be used to test homogeneity of the item discrimination. Infit is sensitive to unexpected patterns in the observations that are appropriate for the subject. It is used to test the assumption of no guessing. For dichotomous items, it has been proven that these statistics are measures of the heterogeneity of item discrimination (Wu & Adams, 2007, pp 85-87). MNSQ fit statistics range from 0 to infinity with an expectation of 1. If they lie outside 95% confidence interval for the expected value of the MNSQ (which under the null hypothesis is 1.0), we reject the null hypothesis that the data conforms to the model. If the MNSQ fit statistic lies outside the interval, then the corresponding T statistics will have an absolute value that exceeds 2.0 warranting closer inspection of associated items (note: ** next to the inft or outfit indicates the misfit).